

Malthus Meets Romer: Population Growth Under Information Frictions

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Abstract

Households disagree about whether larger populations enhance productivity (Romerians) or diminish it (Malthusians). I embed this disagreement into a Barro–Becker fertility model with global games. Wrong beliefs generate different pathologies. In Romerian worlds, Malthusians cause prolonged stagnation through an *informational poverty trap*. In Malthusian worlds, Romerians generate non-monotone dynamics, including boom-bust cycles. Optimal communication policy by the social planner is truthful under a dynastic welfare criterion but shaded under utilitarian ones. Survey data confirm that Malthusian beliefs significantly correlate with lower fertility.

JEL Classification: D83, J13, O40, C72

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1 Introduction

How many children should a family have? The answer depends critically on beliefs about the economic consequences of population size. A parent who believes, in the spirit of [Romer \(1990\)](#), that more people generate more ideas and faster growth might view an additional child as contributing to future prosperity. A parent who fears, in the spirit of [Malthus \(1798\)](#), that more people strain finite resources might view the same child as hastening collective impoverishment. These beliefs directly shape fertility decisions under intergenerational altruism, and the resulting population dynamics feed back into productivity outcomes that either validate or refute the beliefs. This paper studies the consequences of this feedback loop when households hold heterogeneous beliefs about the population-productivity relationship.

To formalize this idea, I develop an overlapping generations model in which households make fertility decisions under uncertainty about a key structural parameter, the scale effect of population on productivity growth. Households observe the same aggregate data but interpret it through different priors. “Romerians” believe that larger populations accelerate innovation and growth, while “Malthusians” believe that larger populations depress living standards. These beliefs are transmitted across generations through a combination of cultural inheritance and Bayesian updating from observed outcomes. Children are costly to raise in time rather than in goods. Following [Becker \(1960\)](#), a parent devotes a fraction $\psi(n)$ of a unit time endowment to child-rearing and supplies the remainder to the labor market. Productivity, in turn, evolves according to a multiplicative log-normal process whose drift depends on population, so that today’s fertility choices shape tomorrow’s productivity through the size of the next generation. Together, these ingredients yield a tractable framework in which beliefs, fertility, and growth co-evolve, and in which the analytical results below follow from a log-linear approximation around a balanced-growth path.

The model rests on a single behavioral premise: beliefs about population’s economic consequences shape fertility. It is worth establishing this premise empirically before drawing out its theoretical implications. Using data from the General Social Survey (GSS), I document robust negative correlations between Malthusian beliefs (measured by agreement with the statement that Earth cannot continue to support population growth) and both ideal and actual fertility. Individuals who hold Malthusian beliefs report ideal fertility approximately 0.27 children lower than non-Malthusians and have 0.29 fewer children, differences of roughly 11 and 16 percent relative to the respective sample means. These relationships persist after controlling for age, education, sex, and survey year, and they are present across demographic groups. The effect on *actual* fertility is at least as large as the effect on stated preferences, suggesting that beliefs translate into consequential behavior rather than cheap

talk. With this empirical foundation in place, I turn to the theoretical analysis of how belief heterogeneity shapes long-run dynamics.

The central theoretical contribution is to show that wrong beliefs generate qualitatively different pathologies depending on which worldview is correct. This asymmetry arises from a feature of Bayesian learning about scale effects: the informativeness of productivity signals depends on population size. When population is near the reference level $\bar{N}$, productivity growth reveals almost nothing about whether scale effects are positive or negative. Learning therefore requires population to deviate substantially from this uninformative region, but such deviations require the very optimistic beliefs that pessimism suppresses. This interaction between beliefs, fertility, and the informativeness of data is the engine driving the paper's main results.

Consider first a Romerian world where scale effects are truly positive. Here Malthusian minorities do not merely slow convergence to efficiency; they can trap the economy in a state of prolonged stagnation. Pessimistic beliefs suppress fertility, keeping population near the reference level $\bar{N}$. A population near $\bar{N}$ generates uninformative signals about the population-productivity relationship, and uninformative signals fail to correct pessimistic beliefs. The trap is self-reinforcing: the economy remains stuck not because Malthusianism is correct, but because population never deviates enough from $\bar{N}$ to reveal that it is wrong. I call this the *informational poverty trap*. While eventual convergence to correct beliefs occurs almost surely, the expected escape time can exceed any relevant historical horizon.

Now consider the mirror case, a Malthusian world where scale effects are truly negative. Here the dynamics differ starkly. Demographic selection always favors optimists, who have more children regardless of whether their optimism is warranted. Bayesian learning, by contrast, favors pessimists when population growth leads to poor outcomes. These two forces pull the Romerian share ϕ_t in opposite directions, and the balance between them itself depends on the state. The paper's second main theoretical result establishes this opposition formally. In any Malthusian world, the expected drift in ϕ_t is strictly positive when population sits at the reference $\bar{N}$ (where signals are uninformative and demographic selection dominates) and strictly negative once population deviates enough that the learning channel overwhelms selection. Monotone convergence is therefore ruled out, and the system inherits the potential for non-monotone dynamics, including the boom-bust cycles familiar from pre-industrial demography. During expansions, optimists outbreed pessimists and population grows. Eventually the economy hits constraints, outcomes deteriorate, learning corrects beliefs toward Malthusianism, and fertility falls. Recovery begins, and the pattern can repeat.

Taken together, the two cases reveal a sharp duality: *in Romerian worlds, Malthusians*

cause stagnation; in Malthusian worlds, Romerians generate non-monotone dynamics, including cycles. Wrong beliefs are always costly, but the nature of the cost depends on the direction of the error.

This duality is more than a theoretical curiosity. It provides a unified lens for interpreting demographic history, though one that admits two competing interpretations of the pre-industrial Malthusian era. The conventional interpretation is that the world was genuinely Malthusian before industrialization: scale effects were negative, and the cycles documented by economic historians (Clark, 2007; Voigtländer and Voth, 2013) reflect the boom-bust dynamics of a true Malthusian regime. Under this view, the Industrial Revolution marked a genuine technological regime change. The alternative interpretation, enabled by the informational poverty trap, is that the world was always Romerian, but humanity was stuck in a trap where population never deviated enough from the reference to generate the information needed to escape pessimism. On this view, the Industrial Revolution was not a technological regime change but an *informational escape*. Some combination of circumstances pushed population into the informative region, beliefs updated toward Romerianism, and the trap dissolved. Both interpretations are consistent with the historical evidence, and distinguishing them has profound implications for understanding the modern fertility decline. The model interprets that decline variously as a partial drift back toward the trap, an efficient response to a renewed Malthusian regime, or a quantity-quality substitution unrelated to scale effects.

If wrong beliefs can trap economies in stagnation or generate damaging cycles, a natural question is whether policy can help, and what “help” even means when the population itself is endogenous to the policy. Addressing this question requires careful attention to welfare evaluation, since, as Golosov et al. (2007) emphasize, standard welfare criteria become problematic when the set of agents depends on policy. I analyze optimal policy under two welfare criteria that span a substantively distinct normative spectrum: a dynastic criterion that respects Barro–Becker preferences, and a total utilitarian criterion that aggregates utility across all individuals who exist under the true law of motion. The two criteria are sufficient to bring out the genuinely distinct policy implications. A dynastic planner is fundamentally an information provider, while a total utilitarian planner is an externality corrector. The former is indifferent to belief correction beyond information provision; the latter has a stake in both truth and population. An average-utilitarian benchmark sits between them and is treated as a corollary.

To keep the policy analysis sharp, I focus on the unique-equilibrium region of the underlying global game, defined by the local contraction condition on the aggregate best response. The threshold I identify is a contraction condition, which delivers *local* uniqueness around a candidate fixed point. The trap region, where the policy intuition has the most bite,

lies inside the contraction region, so the substantive content of the policy analysis is unaffected by this focus. Equilibrium selection in the multiple-equilibria region is an interesting open question that I leave for future work, and I flag explicitly which results depend on the local-uniqueness restriction.

The optimal announcement policy differs substantially across the two criteria. A dynastic planner favors truthful communication, since under subjective preferences any informative signal is welfare-improving and there is no externality to correct. A total utilitarian planner shades announcements toward optimism (in a Romerian world) by an amount that depends on the size of the externality, the deviation of population from the reference level, and the dispersion of priors. An average-utilitarian planner shades less than the total utilitarian but more than the dynastic, because it internalizes the per-capita scale-effect spillover but not the population-size motive. Normative conclusions about the costs of wrong beliefs thus depend critically on the welfare criterion employed, and the contrast across criteria is itself a substantive result of the policy analysis.

Beyond the choice of welfare criterion, the analysis of public information itself reveals a novel asymmetry. In standard coordination games, public information can be harmful by inducing excessive coordination on potentially incorrect signals ([Morris and Shin, 2002](#)). In my setting, the welfare implications depend on where the economy sits. When the economy is trapped in the uninformative region, public information is most valuable, since a credible announcement can coordinate beliefs and push population into the informative region, initiating escape from the trap. But this is precisely when the planner faces maximum uncertainty: the same uninformative signals that trap households also limit what the planner can learn. The planner’s dilemma is sharpest when intervention matters most.

This tension yields a policy prescription that differs from standard results. In the informational poverty trap, even a moderately noisy announcement can improve welfare by breaking the self-reinforcing pessimism, whereas in a Malthusian world with non-monotone dynamics, announcements require higher precision to avoid amplifying volatility. The heterogeneous strategic interaction, with Romerians exhibiting strategic complementarity and Malthusians exhibiting strategic substitutability, provides a natural stabilizer that makes public information less dangerous than in pure-complementarity environments, though it does not eliminate the risk entirely.

Related Literature

This paper connects several strands of literature: endogenous fertility and population, heterogeneous beliefs in macroeconomics, global games, and cultural transmission.

The foundational work on endogenous fertility is [Becker \(1960\)](#), who introduced both the

quantity-quality tradeoff and the time-cost-of-children formulation that I adopt, and [Barro and Becker \(1989\)](#), who embedded fertility choice in a dynastic optimization framework. Subsequent contributions have enriched this framework with human capital accumulation ([Galor and Weil, 2000](#)), mortality transitions ([Kalemlı-Ozcan, 2003](#)), and intergenerational transfers ([Lee, 2003](#)). The unified growth theory of [Galor \(2011\)](#) synthesizes these elements to explain the demographic transition from Malthusian stagnation to modern growth. My contribution differs by introducing belief heterogeneity as a driver of fertility differentials and population dynamics, rather than focusing on the evolution of technology or human capital.

The debate over population scale effects has a long history. [Malthus \(1798\)](#) articulated the view that population growth outstrips resource growth, leading to immiseration. [Romer \(1990\)](#) and [Kremer \(1993\)](#) formalized the opposing view that larger populations generate more ideas, accelerating technological progress. [Jones \(1995\)](#) and [Jones \(2022\)](#) provide frameworks that nest both perspectives. Empirical work by [Ashraf and Galor \(2013\)](#) finds evidence for positive scale effects over very long horizons, while [Voigtländer and Voth \(2013\)](#) documents Malthusian dynamics in pre-industrial Europe. I do not take a stand on which view is correct; rather, I study the consequences of households holding different views.

The literature on heterogeneous beliefs has grown substantially. In macroeconomics, [Angeletos and La’O \(2013\)](#) and [Angeletos et al. \(2018\)](#) examine how dispersed information affects business cycles and sentiments. [Bordalo et al. \(2018\)](#) develop diagnostic expectations with belief overreaction. [Malmendier and Nagel \(2011\)](#) document that personal experiences shape macroeconomic expectations. In finance, [Simsek \(2013\)](#), [Geanakoplos \(2010\)](#), and [Scheinkman and Xiong \(2003\)](#) study asset pricing under disagreement. My paper differs by modeling disagreement about structural parameters governing long-run relationships rather than beliefs about current states or short-run forecasts.

The global games literature, pioneered by [Carlsson and Van Damme \(1993\)](#) and developed by [Morris and Shin \(1998\)](#) and [Morris and Shin \(2003\)](#), provides tools for analyzing games with strategic complementarities under incomplete information. I adapt these techniques to a setting with both strategic complementarities (among Romerians) and strategic substitutabilities (among Malthusians). This combination creates novel analytical challenges but also generates richer dynamics than standard global games applications such as currency crises ([Morris and Shin, 1998](#)) or bank runs ([Goldstein and Pauzner, 2005](#)). Because standard global-games selection theorems do not directly apply to a continuous-action, heterogeneous-strategic-interaction game with two-dimensional belief uncertainty, I focus the policy analysis on the local-uniqueness region of the underlying game.

The literature on cultural transmission, developed by [Cavalli-Sforza and Feldman \(1981\)](#), [Bisin and Verdier \(2001\)](#), and [Bisin and Verdier \(2011\)](#), provides frameworks for understand-

ing how preferences and beliefs persist across generations. [Doepke and Zilibotti \(2019\)](#) apply these tools to study parenting styles. I incorporate cultural transmission as one channel through which beliefs evolve, alongside Bayesian learning from observed outcomes.

The analysis of public information in coordination games relates to work by [Morris and Shin \(2002\)](#) on the social value of public information. They show that in games with strategic complementarities, public information can be harmful if it crowds out private information. In my setting, the welfare implications of public announcements are more nuanced because of the heterogeneous strategic interaction and the state-dependent value of coordination: public information is most valuable in the trap, where it is also most uncertain.

An important related literature addresses welfare evaluation with endogenous population. [Goloso et al. \(2007\)](#) provide a systematic analysis of efficiency concepts when fertility is a choice variable, showing that standard welfare theorems fail under total or average utilitarianism but that a modified Pareto criterion can restore efficiency of competitive equilibria. [Blackorby et al. \(2005\)](#) develop the axiomatic foundations of population ethics and characterize critical-level utilitarianism as a resolution to the “Repugnant Conclusion” of [Parfit \(1984\)](#). I build on these insights by analyzing how dynastic and total utilitarian welfare criteria affect optimal communication policy in a setting with belief heterogeneity.

Finally, my paper relates to work on poverty traps and multiple equilibria in development economics. [Azariadis and Drazen \(1990\)](#) formalize poverty traps arising from threshold externalities. [Banerjee and Newman \(1993\)](#) study occupational choice with wealth constraints. The informational poverty trap I identify differs from these mechanisms: the trap arises not from technological nonconvexities or credit constraints but from the endogenous uninformative-ness of signals when the economy remains in a particular region of the state space.

The paper proceeds as follows. Section 2 presents the model environment. Section 3 presents empirical evidence validating the model’s central prediction that Malthusian beliefs reduce fertility. Section 4 analyzes strategic complementarity and substitutability and isolates the local-uniqueness region. Section 5 examines long-run dynamics, including the informational poverty trap and the non-monotone dynamics of Malthusian worlds. Section 6 discusses the alignment between theoretical predictions and demographic history. Section 7 analyzes the value of public information and characterizes optimal communication policy. Section 8 concludes. All proofs and auxiliary results are collected in the Appendix.

2 Model

I develop a discrete-time overlapping generations model with dynastic altruism. The economy is populated by a continuum of households who make fertility decisions. Households

differ in their beliefs about the structural relationship between population and productivity growth.

2.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \dots$. Each period, a generation of adults makes economic decisions; their children become adults in the following period.

Time Endowment and Cost of Children. Each adult is endowed with one unit of time. An adult who chooses fertility n_{it} spends a fraction $\psi(n_{it}) \in [0, 1)$ of that time on child-rearing and supplies the remainder $1 - \psi(n_{it})$ to the labor market. Child-rearing therefore competes directly with market work for the household's time, and the implicit price of an additional child is the forgone labor income. This time-cost specification, originally articulated by [Becker \(1960\)](#), has the important property that the resource constraint on consumption is multiplicative rather than additive. A household's consumption is its labor income net of the time spent on children, not its labor income minus a goods expenditure. The multiplicative structure plays a central analytical role in what follows by making the household value function tractable to a first-order log-linear approximation.

Production. A single final good is produced using effective labor according to the linear technology

$$Y_t = A_t L_t, \quad L_t = N_t \cdot \int_0^1 (1 - \psi(n_{it})) di, \quad (1)$$

where Y_t is output, A_t is total factor productivity (TFP), N_t is the adult population, and L_t is aggregate effective labor supply. I interpret A_t broadly as an “effective” TFP that incorporates any potential dilution effects from fixed factors. The competitive wage equals the marginal product of effective labor:

$$w_t = A_t. \quad (2)$$

A type- j household with fertility $n_{j,t}$ supplies $1 - \psi(n_{j,t})$ units of labor and earns $A_t(1 - \psi(n_{j,t}))$. With no other income or expenditure, consumption is

$$c_{j,t} = A_t(1 - \psi(n_{j,t})). \quad (3)$$

Aggregate per-capita output and consumption coincide automatically.

TFP Dynamics: Multiplicative Log-Normal Specification. TFP evolves according to

$$A_{t+1} = A_t \cdot \exp(\bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}), \quad \eta_{t+1} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad (4)$$

where $\bar{g} \geq 0$ is baseline log-productivity growth, $\theta^* \in \mathbb{R}$ is the *scale-effect parameter*, $\bar{N} > 0$ is an empirically anchored *reference population*, and η_{t+1} is an i.i.d. structural productivity shock independent of all histories. The multiplicative log-normal specification has three substantial advantages over the additive specification used in earlier formulations of this paper. First, A_{t+1} is automatically positive for any realization of η_{t+1} , so the log-utility framework is well defined throughout the state space. Second, taking logs gives the exact identity

$$\ln A_{t+1} = \ln A_t + \bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}, \quad (5)$$

which is linear in $\ln A_t$, $\ln(N_t/\bar{N})$, and η_{t+1} . The technology side of the model contributes *no* Taylor remainder; any approximation in what follows enters only through the optimal fertility policy and the value-function conjecture below. Third, the reference population $\bar{N}$ appears explicitly, so the model is invariant under rescaling of the population unit: only $\ln(N_t/\bar{N})$ matters for the dynamics, not the absolute level of N_t . Throughout, I refer to $g(N_t; \theta) \equiv \bar{g} + \theta \ln(N_t/\bar{N})$ as the productivity-growth function.

When $\theta^* > 0$, larger populations generate faster productivity growth (Romer); when $\theta^* < 0$, larger populations slow productivity growth (Malthusian dynamic congestion). When $N_t = \bar{N}$, the deterministic component of productivity growth equals $\bar{g}$ regardless of θ^* . Observations near $N_t = \bar{N}$ are therefore *uninformative* about the scale-effect parameter, even though they remain noisy through η_{t+1} . This feature is central to the informational poverty trap.

Cost-dilution interpretation. The parameter $\theta < 0$ in this model represents *dynamic* congestion or environmental drag, namely the marginal effect of population on the growth rate of productivity. It does not capture classical Malthusian fixed-factor dilution operating on the level of output. Capturing the latter would require an additional production margin (a fixed factor in the production function), which is straightforward but distinct from the channel modeled here. The current setup is best understood as a reduced-form representation of dynamic effects of population on innovation, in the spirit of [Romer \(1990\)](#) and [Jones \(1995\)](#) on the positive- θ side, and of environmental-drag and congestion models on the negative- θ side.

Households and Beliefs. The economy contains a unit mass of dynastic households indexed by $i \in [0, 1]$. Each household holds a belief θ_i about the true value of the scale-effect parameter.

Assumption 1 (Belief Types). *A fraction $\phi_t \in [0, 1]$ of households are Romerians with belief $\theta_R > 0$; the remaining $1 - \phi_t$ are Malthusians with belief $\theta_M < 0$.*

A “Romerian world” has $\theta^* > 0$; a “Malthusian world” has $\theta^* < 0$. Households do not know θ^* with certainty but hold a common prior $\pi_t \in [0, 1]$ that the world is Romerian and update via Bayes’ rule. The distribution of beliefs ϕ_t is common knowledge in each period, but individual household types are private information.

Time-Cost Function. I impose the following regularity conditions.

Assumption 2 (Time Cost of Children). *The time-cost function $\psi : \mathbb{R}_+ \rightarrow [0, 1]$ satisfies (i) $\psi(0) = 0$; (ii) $\psi'(n) > 0$ for $n > 0$ and $\psi''(n) \geq 0$ for all $n \geq 0$; (iii) $\psi(n) < 1$ for finite n with $\lim_{n \rightarrow \infty} \psi(n) = 1$; and (iv) the marginal time cost $\psi'(n)/(1 - \psi(n))$ is strictly increasing in n with $\lim_{n \rightarrow \infty} \psi'(n)/(1 - \psi(n)) = \infty$.*

The boundary condition (ii) is now stated as $\psi'(n) > 0$ only for $n > 0$ (rather than for all $n \geq 0$), so that the working specification $\psi(n) = 1 - e^{-\gamma n^2/2}$ introduced below satisfies the assumption literally. The crucial regularity for first-order conditions is the strict monotonicity of the effective marginal time cost $\psi'(n)/(1 - \psi(n))$ in (iv), which holds at all $n \geq 0$ under the working specification.

For analytical tractability, I use the exponential–quadratic specification

$$\psi(n) = 1 - e^{-\frac{\gamma}{2}n^2}, \quad \gamma > 0, \quad (6)$$

which yields the convenient identity $\psi'(n)/(1 - \psi(n)) = \gamma n$. This identity exactly mirrors the linear marginal-cost structure of a goods-cost specification while keeping consumption bounded between zero and A_t .

Preferences. Households have dynastic preferences:

$$U_{it} = u(c_{it}) + \beta n_{it} \mathbb{E}_{it} [V_{i,t+1}], \quad (7)$$

where $u(c) = \ln c$, $\beta \in (0, 1)$ is the discount factor, and $V_{i,t+1}$ is the continuation value for the dynasty starting next period. Throughout, I assume equilibrium fertility satisfies $\beta n_j^* < 1$

so that the dynastic value function is well defined. Substituting $c_{it} = A_t(1 - \psi(n_{it}))$ gives the additive decomposition

$$\ln c_{it} = \ln A_t + \ln(1 - \psi(n_{it})). \quad (8)$$

The first term depends only on aggregate productivity, while the second depends only on own fertility through the time cost. This separation is the analytical advantage of the time-cost specification: $\ln A_t$ carries dynamic, aggregate information, while $\ln(1 - \psi(n_{it}))$ carries the static individual choice.

Population Dynamics. Aggregate population evolves according to

$$N_{t+1} = \bar{n}_t \cdot N_t, \quad \bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}, \quad (9)$$

where $n_{R,t}$ and $n_{M,t}$ denote the fertility choices of Romerians and Malthusians, respectively.

2.2 Belief Formation and Intergenerational Transmission

Beliefs evolve through cultural transmission and Bayesian learning.

Cultural Transmission. Children initially adopt their parents' beliefs with probability $\lambda \in [0, 1]$ and learn from observed outcomes with probability $1 - \lambda$.

Assumption 3 (Cultural Transmission). *A child born to a type- j household adopts type j with probability λ and forms beliefs through learning with probability $1 - \lambda$.*

When $\lambda = 1$ beliefs are perfectly persistent across generations; when $\lambda = 0$ beliefs are formed entirely through learning.

Bayesian Learning. Children who do not inherit parental beliefs form beliefs by Bayes' rule. Under the multiplicative log-normal TFP law of motion (4), observed log-productivity growth is

$$s_t \equiv \ln(A_{t+1}/A_t) = \bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}, \quad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2). \quad (10)$$

This signal is observed exactly without further measurement noise; the only source of randomness in the signal is the structural shock η_{t+1} that already appears in the law of motion.

Given s_t and N_t , the likelihood under belief θ is

$$\ell(s_t \mid \theta, N_t) = \frac{1}{\sqrt{2\pi}\sigma_\eta} \exp\left(-\frac{(s_t - \bar{g} - \theta \ln(N_t/\bar{N}))^2}{2\sigma_\eta^2}\right), \quad (11)$$

and Bayes' rule gives the posterior

$$\pi_{t+1} = \frac{\pi_t \cdot \ell(s_t \mid \theta_R, N_t)}{\pi_t \cdot \ell(s_t \mid \theta_R, N_t) + (1 - \pi_t) \cdot \ell(s_t \mid \theta_M, N_t)}. \quad (12)$$

Children who learn adopt Romerian beliefs with probability π_{t+1} and Malthusian beliefs with probability $1 - \pi_{t+1}$.

Signal Informativeness. The Kullback–Leibler divergence between the signal distributions under the two beliefs is

$$\mathcal{I}_t = \frac{(\theta_R - \theta_M)^2 [\ln(N_t/\bar{N})]^2}{2\sigma_\eta^2}. \quad (13)$$

When $N_t = \bar{N}$, $\mathcal{I}_t = 0$: the signal is completely uninformative about θ^* . As $|\ln(N_t/\bar{N})|$ grows, the signal becomes increasingly informative. This dependence of informativeness on the deviation of population from its reference is the source of the informational poverty trap and, in Malthusian worlds, of the opposing forces analyzed in Section 5.

Evolution of Belief Distribution. Combining cultural transmission and learning, the Romerian share evolves as

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1 - \lambda) \pi_{t+1}, \quad (14)$$

where $\phi_t^{birth} = \phi_t n_{R,t}/\bar{n}_t$ is the fraction of children born to Romerians. The Romerian share is pushed upward by differential fertility (when $n_R > n_M$, so $\phi_t^{birth} > \phi_t$) and by learning (when evidence favors the Romerian model, so $\pi_{t+1} > \pi_t$).

2.3 Equilibrium

Definition 1 (Equilibrium). *An equilibrium consists of fertility policies $n_R(A_t, N_t, \phi_t, \pi_t)$, $n_M(A_t, N_t, \phi_t, \pi_t)$; aggregate fertility $\bar{n}(A_t, N_t, \phi_t, \pi_t)$; and laws of motion for $(A_{t+1}, N_{t+1}, \phi_{t+1}, \pi_{t+1})$ given by (4), (9), (14), and (12), such that (a) each household's fertility choice maximizes (7) subject to (3) given subjective beliefs and the aggregate fertility function; (b) expectations about aggregate fertility are consistent with the equilibrium aggregate; and (c) the actual laws of motion use the true θ^* and realized η_{t+1} .*

Two features merit emphasis. First, households optimize under their subjective beliefs but the actual economy evolves according to θ^* . This allows for systematic forecast errors. Second, the state vector includes the posterior π_t explicitly, so the Bellman equation is genuinely Markov. The decentralized allocation does not internalize the externality that population growth imposes on TFP through (4). Even with common correct beliefs, the resulting allocation is not the social planner's optimum.

Remark 1 (Type-switching and the Bellman). *The Bellman equation below treats a type- j parent's continuation value as the value function of a type- j child, even though the child may switch with probability $1 - \lambda$. An exact formulation would replace $\mathbb{E}_j[V_j]$ by $\lambda \mathbb{E}_j[V_j] + (1 - \lambda) \mathbb{E}_j[V_{\text{learn}}]$, where V_{learn} is a type-mixture continuation expected by the parent. The two formulations coincide when λ is close to 1 or when V_R and V_M are close in the relevant region; both conditions hold near the steady states studied below. I use the simpler formulation in the main text and discuss the more general case in Appendix A.*

2.4 Household Optimization

The Bellman for a type- j household reads:

$$V_j(A, N, \phi, \pi) = \max_{n \geq 0} \left\{ \ln A + \ln(1 - \psi(n)) + \beta n \cdot \mathbb{E}_j[V_j(A', N', \phi', \pi')] \right\}. \quad (15)$$

The choice variable n enters only through the static term $\ln(1 - \psi(n))$ and the multiplier βn on the continuation value. The aggregate state (A', N', ϕ', π') is, from the perspective of the atomistic household, exogenous to its individual choice.

Local log-linear approximation. I work with a first-order log-linear approximation of V_j around a balanced-growth reference point. Specifically, I conjecture

$$V_j(A, N, \phi, \pi) \approx v_j(\phi, \pi) + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}), \quad (16)$$

where the coefficients $(\xi_{A,j}, \xi_{N,j})$ and the residual $v_j(\phi, \pi)$ are determined by coefficient matching at the reference point. The technology side of the Bellman is exactly linear in $(\ln A, \ln(N/\bar{N}))$ under the multiplicative log-normal specification, so any deviation from (16) comes from the dependence of the optimal fertility n_j^* on the state through the FOC (17) below. Treating n_j^* as locally constant in $(\ln A, \ln(N/\bar{N}))$ at the reference point yields the closed-form coefficients reported in Lemma 1. The qualitative results (signs of strategic interaction, the informational poverty trap, and so on) depend only on the sign of $\xi_{N,j}$, which

inherits the sign of θ_j under any smooth value function.¹

The first-order condition for fertility is:

$$\frac{\psi'(n_j)}{1 - \psi(n_j)} = \beta \mathbb{E}_j[V_j(A', N', \phi', \pi')]. \quad (17)$$

Under (6), this simplifies to $\gamma n_j = \beta \mathbb{E}_j[V_j]$. The FOC itself is exact in n_j given the right-hand side; the local-approximation caveat enters only through the value-function conjecture (16) that the right-hand side depends on.

Lemma 1 (Equilibrium Fertility, Local Coefficients). *Under Assumptions 1–3 and the local log-linear conjecture (16), the value function coefficients evaluated at the balanced-growth reference are*

$$\xi_{A,j} = \frac{1}{1 - \beta n_j^*}, \quad \xi_{N,j} = \frac{\beta n_j^* \theta_j}{(1 - \beta n_j^*)^2}. \quad (18)$$

Equilibrium fertility satisfies a fixed-point equation in n_j^ and can be written as $n_j^* = n^* h(\theta_j; \bar{n})$ with h increasing in θ . Moreover, $n_R^* > n_M^*$ for all equilibrium values of $\bar{n}$.*

The proof, including a careful statement of the local nature of (18), is in Appendix A. The economic insight is that households who believe population enhances productivity (Romerians) perceive a higher return to children and choose higher fertility. This testable prediction motivates the empirical analysis in the next section.

3 Empirical Evidence: Beliefs and Fertility

A central prediction of the model is that Malthusian beliefs reduce fertility. Lemma 1 establishes that households believing population diminishes productivity choose lower fertility than households believing population enhances it. This section examines empirical support using survey data on beliefs and fertility.

3.1 Data

I use data from the General Social Survey (GSS), a nationally representative survey of U.S. adults conducted by NORC at the University of Chicago. The key belief variable comes from the question: “The earth cannot continue to support population growth at its present rate.”

¹Reading (16) as a first-order log-linear approximation around a balanced-growth reference allows for higher-order remainder terms in the state. The closed-form expressions for $(\xi_{A,j}, \xi_{N,j})$ below should be interpreted as their values at the reference point, not as global slopes. The qualitative conclusions of the paper do not require global exactness; they require only the local sign structure, which is robust under standard regularity.

Respondents indicate agreement on a five-point scale. I classify those who agree or strongly agree as holding *Malthusian beliefs*. This question was asked in the 2000 and 2010 waves.

I examine two fertility measures: stated preferences (“ideal number of children”) and realized fertility (actual number of children). Examining both lets me test whether beliefs shape intentions and outcomes. The sample contains 1,822 observations with valid responses to the belief and fertility questions and to demographic controls.

3.2 Descriptive Patterns

Table 1 presents summary statistics. The mean ideal number of children is 2.50 (s.d. 0.87); the mean actual number is 1.81 (s.d. 1.64). Approximately 55 percent of respondents hold Malthusian beliefs.

Table 1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Ideal number of children	1,822	2.50	0.87	0	7
Actual number of children	1,822	1.81	1.64	0	8
Malthusian belief (binary)	1,822	0.55	0.50	0	1
Malthusian belief (1–5 scale)	1,822	3.44	1.03	1	5
Age	1,822	46.8	17.3	18	89
Years of education	1,822	13.3	2.96	0	20
Female	1,822	0.56	0.50	0	1

Notes: Data from General Social Survey, 2000 and 2010 waves. Malthusian belief (binary) equals one if respondent agrees or strongly agrees that Earth cannot continue to support population growth.

Figure 1 displays mean ideal and actual fertility by Malthusian belief intensity. There is a clear monotonic negative relationship between Malthusian belief strength and both fertility measures. Respondents who strongly disagree report an average ideal of 2.98 children and have 1.76 children, while those who strongly agree report an average ideal of 2.24 children and have 1.52 children.

A potential concern is that this relationship is driven by education. Figure 2 addresses this concern by displaying mean ideal fertility by education group and belief type. Within every education category, Malthusians report lower ideal fertility than non-Malthusians. The gap ranges from 0.20 children among college graduates to 0.45 children among those with less than high school education.

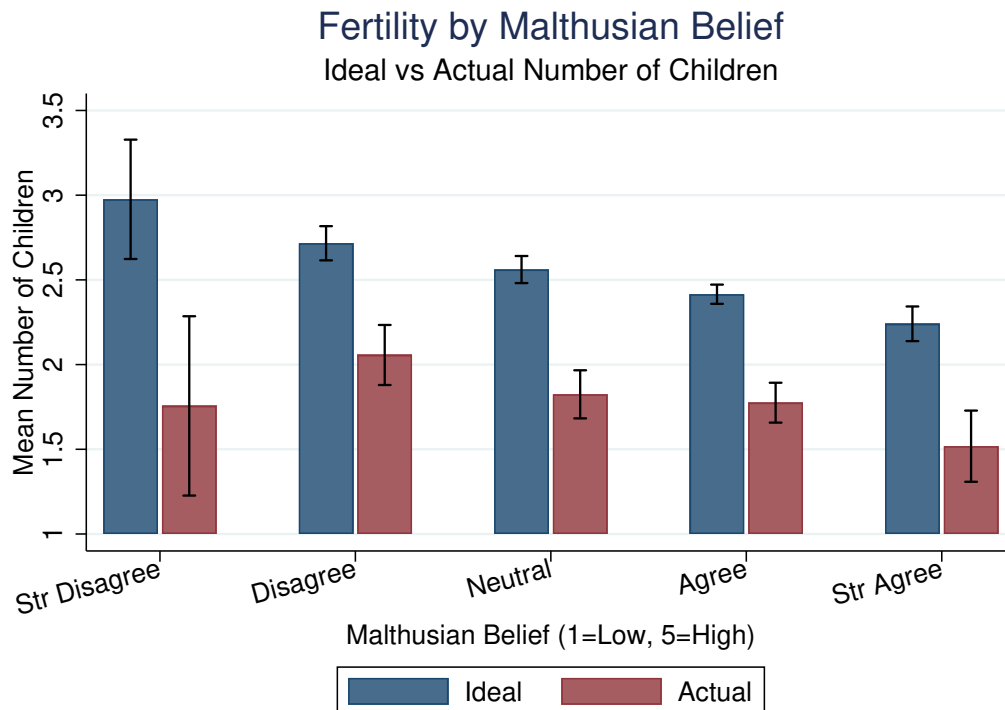


Figure 1. Fertility by Malthusian Belief Intensity

Notes: Data from General Social Survey, 2000 and 2010 waves (N = 1,822). Bars show mean ideal (navy) and actual (maroon) number of children for each level of the Malthusian belief scale. Error bars indicate 95% confidence intervals.

3.3 Regression Analysis

Table 2 presents linear regressions of both ideal and actual fertility on Malthusian beliefs with demographic controls. All specifications include both age and age squared, so the coefficient on age in the table reflects the linear component only. The joint effect of age in any given year requires combining the linear and quadratic terms.

Malthusians report ideal fertility 0.27 children lower than non-Malthusians (column 1), or about 11 percent of the sample mean. The coefficient on actual fertility is -0.29 (column 3), implying Malthusians have approximately 16 percent fewer children than non-Malthusians. The coefficient on actual fertility is *larger* in absolute value than the coefficient on ideal fertility, suggesting that beliefs translate into behavior with at least as much force as they affect stated preferences.

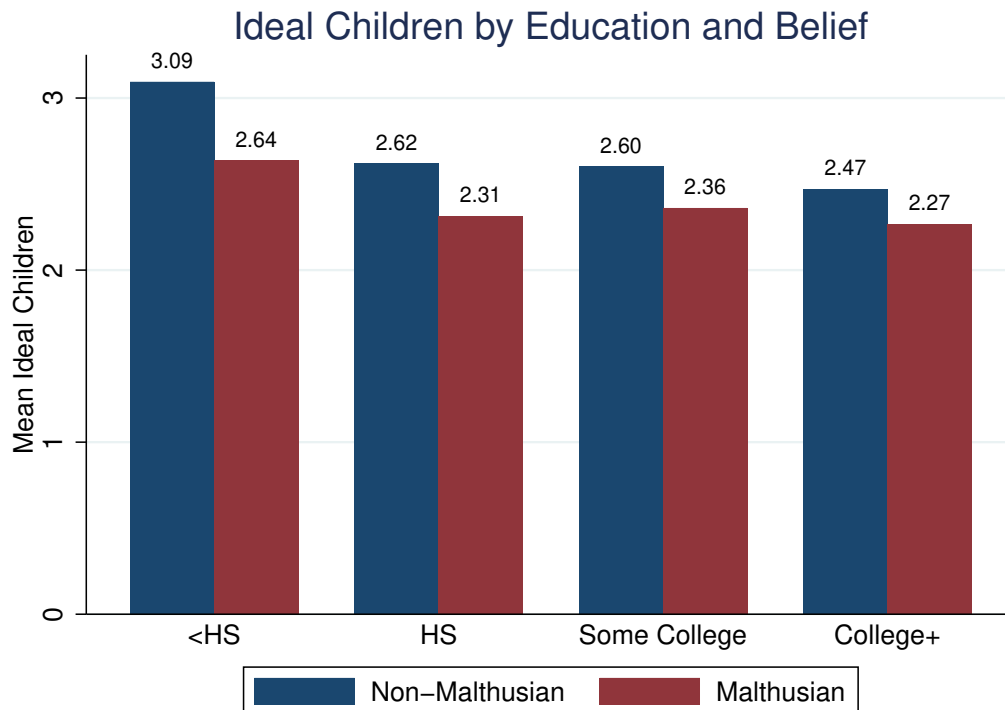


Figure 2. Ideal Number of Children by Education and Malthusian Belief

Notes: Bars show mean ideal number of children for non-Malthusians (navy) and Malthusians (maroon) within each education category.

3.4 Completed Fertility and Heterogeneity

A concern with actual fertility is that younger respondents may not have completed child-bearing. Table 3 restricts the sample to women aged 45+ and men aged 50+. Among those with completed fertility, Malthusians have 0.26 fewer children than non-Malthusians, similar to the full-sample estimate. The relationship also appears among those who have not completed fertility, suggesting belief-driven differentials emerge early in the reproductive life course.

Table 4 examines heterogeneity by education. The belief-fertility relationship is present across education groups for both outcome measures. For ideal fertility, the coefficient is somewhat larger for less educated respondents. For actual fertility the pattern reverses, with the coefficient larger among the more educated. This is suggestive rather than dispositive, since I do not test the cross-group difference formally. With that caveat, the descriptive pattern is consistent with the interpretation that Malthusian beliefs affect fertility preferences across the education distribution but translate into behavior more strongly among the more educated, perhaps because they face fewer constraints on achieving their fertility targets.

Table 2. Effect of Malthusian Beliefs on Fertility

	Ideal Fertility		Actual Fertility	
	(1)	(2)	(3)	(4)
Malthusian	−0.268*** (0.041)		−0.292*** (0.066)	
Malthusian (continuous)		−0.153*** (0.021)		−0.146*** (0.033)
Age	−0.014** (0.006)	−0.015** (0.006)	0.112*** (0.010)	0.111*** (0.010)
Female	0.015 (0.040)	0.026 (0.040)	0.091 (0.068)	0.104 (0.068)
Education	−0.047*** (0.007)	−0.046*** (0.007)	−0.144*** (0.013)	−0.143*** (0.013)
Year FE	Yes	Yes	Yes	Yes
Observations	1,822	1,822	1,822	1,822
R^2	0.060	0.069	0.262	0.262

Notes: All models include age squared and year fixed effects. Robust standard errors in parentheses. The reported age coefficient is the linear term only; the marginal effect of age combines the linear and quadratic components and is therefore age-specific. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3. Actual Fertility by Completed Fertility Status

	Full Sample (1)	Completed Fertility (2)	Not Yet Completed (3)
Malthusian	−0.292*** (0.066)	−0.257** (0.109)	−0.288*** (0.079)
Observations	1,822	848	974
R^2	0.262	0.112	0.266

Notes: Dependent variable is actual number of children. All models control for age, age squared, sex, education, and year. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5 Interpretation and Limitations

The empirical findings are consistent with the model’s central prediction: individuals who believe population growth is unsustainable desire and have fewer children. The relationships are economically meaningful (gaps of 11 and 16 percent of the respective sample means) and statistically robust.

Two features of the results merit emphasis. First, the relationship holds for *actual* fertil-

Table 4. Heterogeneity in the Belief-Fertility Relationship

	Ideal Fertility		Actual Fertility	
	HS or Less	Some College+	HS or Less	Some College+
Malthusian	−0.318*** (0.069)	−0.235*** (0.049)	−0.209* (0.115)	−0.392*** (0.081)
Observations	821	1,001	821	1,001
R^2	0.045	0.029	0.163	0.240

Notes: Dependent variable is ideal (columns 1–2) or actual (columns 3–4) number of children. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ity, not just stated preferences. This addresses a common concern that surveys capture cheap talk rather than consequential behavior. Second, the coefficient on actual fertility is at least as large as the coefficient on ideal fertility. This pattern suggests either that beliefs operate partly through channels beyond conscious intention, or that individuals with stronger beliefs are more committed to acting on them.

Several limitations warrant emphasis. The evidence remains correlational; it could reflect reverse causality or omitted variable bias. The cross-sectional nature of the data precludes tracking how belief changes affect subsequent fertility decisions. The belief measure also captures concern about global population sustainability, not specifically the scale-effect parameter θ . Despite these limitations, the evidence provides suggestive support for the belief channel emphasized in the model. With this empirical foundation, I now turn to the strategic implications.

4 Strategic Complementarity and Substitutability

This section analyzes strategic interaction in fertility choices. I show that Romerians exhibit strategic complementarity while Malthusians exhibit strategic substitutability, and isolate a local-uniqueness region (defined by a contraction condition on the aggregate best response) within which the policy analysis of Section 7 is conducted.

4.1 The Fertility Game

Consider the fertility choice as a game among households. Each household i chooses n_i to maximize expected utility. Aggregate fertility is $\bar{n} = \int n_i di$. The household's payoff depends on $(n_i, \bar{n})$ through a single channel: higher $\bar{n}$ raises $N' = \bar{n}N$, which through (4) affects A'' and subsequent productivity. Formally, let $U_j(n_j, \bar{n})$ denote the expected utility

of a type- j household choosing n_j when aggregate fertility is $\bar{n}$, with the expectation taken under belief θ_j . Type j exhibits *strategic complementarity* if $\partial^2 U_j / \partial n_j \partial \bar{n} > 0$ and *strategic substitutability* if the cross-partial is negative.

4.2 Best Responses and the Sign of Strategic Interaction

Let $n_j(\bar{n}^e)$ be the fertility choice of type j when expected aggregate fertility is $\bar{n}^e$. The actual aggregate is $\bar{n} = \phi n_R(\bar{n}^e) + (1 - \phi) n_M(\bar{n}^e)$, and rational expectations require $\bar{n}^e = \bar{n}$. Each household knows its own type but is uncertain about other households' realizations and hence about aggregate fertility.

Under the local log-linear approximation, the marginal effect of expected aggregate fertility on the continuation value is

$$\frac{\partial \mathbb{E}_j[V_j]}{\partial \bar{n}^e} = \frac{\xi_{N,j}}{\bar{n}^e}.$$

From (18), the sign of $\xi_{N,j}$ matches the sign of θ_j : $\xi_{N,M} < 0$ for Malthusians and $\xi_{N,R} > 0$ for Romerians. The household FOC has a strictly increasing left-hand side in n_j , so the sign of $dn_j^*/d\bar{n}^e$ coincides with the sign of $\xi_{N,j}$. The sign result is robust beyond the local approximation, since $\xi_{N,j}$ inherits the sign of θ_j under any smooth value function.

Proposition 1 (Strategic Interaction). *Under Assumptions 1–3: (i) Romerians exhibit strategic complementarity, $dn_R^*/d\bar{n}^e > 0$; (ii) Malthusians exhibit strategic substitutability, $dn_M^*/d\bar{n}^e < 0$.*

The proof is in Appendix A. The intuition is straightforward. A Romerian who expects higher aggregate fertility expects higher future population, which under Romerian beliefs implies faster productivity growth and a higher continuation value of each child. The Romerian responds by having more children. A Malthusian who expects higher aggregate fertility expects slower future productivity growth, which lowers the continuation value and reduces fertility.

4.3 Local Uniqueness and the Multiple-Equilibria Region

Define the aggregate best-response slope

$$\rho(\phi) \equiv \phi \frac{dn_R^*}{d\bar{n}^e} + (1 - \phi) \frac{dn_M^*}{d\bar{n}^e}. \quad (19)$$

This quantity is positive when ϕ is sufficiently large (Romerian dominance), negative when ϕ is sufficiently small (Malthusian dominance), and zero at the critical share

$$\phi^* = \frac{|dn_M^*/d\bar{n}^e|}{|dn_R^*/d\bar{n}^e| + |dn_M^*/d\bar{n}^e|}. \quad (20)$$

Strictly speaking, ϕ^* is evaluated locally at the equilibrium $\bar{n}^e$: the derivatives $dn_j^*/d\bar{n}^e$ depend on $\bar{n}^e$ through $n_j^*(\bar{n}^e)$, so ϕ^* is a state-dependent quantity. The qualitative claims that follow concern the local sign of $\rho(\phi)$ at the equilibrium of interest, and the local-uniqueness condition in Proposition 2 should be read as a local condition on the relevant fixed point.

Proposition 2 (Local Uniqueness via Contraction). *Under the conditions of Proposition 1: (i) the contraction condition $|\rho(\phi)| < 1$ is sufficient for the aggregate best response $BR(\bar{n}^e)$ to admit a locally unique fixed point in a neighborhood of the relevant $\bar{n}^e$; (ii) when $\rho(\phi) > 1$, $BR(\bar{n}^e)$ can intersect the 45-degree line transversally at multiple points, generating multiple equilibria that can be locally ordered by aggregate fertility.*

The proof is in Appendix A. The condition $|\rho(\phi)| < 1$ delivers *local* uniqueness via a contraction-mapping argument. In the multiplicity region, equilibrium selection is non-trivial. Standard global-games selection theorems do not directly apply to the present setting, in which actions are continuous, strategic interaction has opposite signs across types, and beliefs lie in a two-dimensional space. A heuristic selection argument is sketched in Appendix A (Proposition 7) and is independent of the policy analysis.

The economic content is meaningful: societies dominated by Romerians may be subject to self-fulfilling fluctuations in fertility, whereas societies dominated by Malthusians have more stable fertility dynamics in the local-uniqueness sense.

Focus on the local-uniqueness region. The trap region of Section 5 typically features beliefs concentrated on $\theta_M < 0$, which weakens or reverses strategic complementarity, so $|\rho(\phi)|$ is small there. The same is true at the interior steady state of the Malthusian world studied in Section 5, where the mix of types keeps ρ bounded. The local-uniqueness region is therefore precisely the region where the policy intuition has the most bite, and I make this explicit by maintaining the following condition throughout the policy analysis.

Assumption 4 (Local-Uniqueness Region). *The policy analysis in Section 7 is conducted under the maintained assumption that $|\rho(\phi_t)| < 1$ at the relevant fixed point for all relevant t .*

Equilibrium selection in the multiple-equilibria region is an interesting open question that I leave for future work, and I flag explicitly which results depend on Assumption 4.²

5 Long-Run Dynamics and the Role of Wrong Beliefs

I now analyze the long-run dynamics of beliefs and population. Wrong beliefs generate qualitatively different pathologies depending on which worldview is correct. In Romerian worlds, Malthusian minorities cause prolonged stagnation through an informational poverty trap. In Malthusian worlds, Romerian minorities generate non-monotone dynamics, including the boom-bust cycles familiar from pre-industrial demography. This asymmetry arises from the interaction between demographic selection (which always favors optimists) and Bayesian learning (which favors correct beliefs, but only when signals are informative).

5.1 Two Forces: Demographic Selection and Learning

The Romerian share ϕ_t evolves according to (14). Two forces drive its evolution.

Demographic selection. Since $n_R > n_M$, $\phi_t^{birth} > \phi_t$ for any interior ϕ_t . Demographic selection pushes ϕ_{t+1} upward regardless of which belief is correct.

Bayesian learning. The direction of π_{t+1} depends on the true θ^* . If $\theta^* > 0$, high population tends to generate high productivity growth, evidence for the Romerian model; if $\theta^* < 0$, high population tends to generate low productivity growth, evidence for the Malthusian model. The *strength* of learning depends on population through the signal informativeness $\mathcal{I}_t$ in (13). When $N_t \approx \bar{N}$, $\mathcal{I}_t \approx 0$ and learning is weak regardless of the truth.

The interaction between these two forces produces sharply different dynamics in Romerian and Malthusian worlds. In Romerian worlds, both forces align toward correct beliefs, and the only question is how quickly correct beliefs spread. The answer can be “arbitrarily slowly” when the economy starts in the trap. In Malthusian worlds, the two forces conflict, and the resulting tension generates the non-monotone dynamics that the rest of this section makes precise.

²For completeness, Appendix A (Proposition 7) sketches a heuristic global-games selection argument. Suppose households observe a private signal $\phi_i = \phi + \zeta_i$ about the Romerian share. As private noise vanishes, threshold strategies appear to select a unique equilibrium: Romerians coordinate on high fertility above a common threshold, while Malthusians anti-coordinate. A fully rigorous selection theorem in this heterogeneous-strategic-interaction setting would require an analog of Frankel et al. (2003).

5.2 Dynamics in a Romerian World: Convergence and the Trap

Theorem 1 (Convergence in Romerian Worlds). *Suppose $\theta^* > 0$ and that the population trajectory remains within the region where (4) is well-defined. Starting from any interior initial condition $(\phi_0, N_0, A_0, \pi_0)$: (i) $\pi_t \rightarrow 1$ almost surely; (ii) $\phi_t \rightarrow 1$ almost surely; (iii) aggregate fertility converges to n_R^* ; (iv) $\sup_t \ln(N_t/\bar{N}) = +\infty$ almost surely.*

The proof is in Appendix A. Both demographic selection and learning favor Romerians when the world is truly Romerian. Demographic selection favors them because they have more children, and learning favors them because observed outcomes confirm their beliefs.

A note on bounded population. The result that $\sup_t \ln(N_t/\bar{N}) = +\infty$ presumes the absence of endogenous bounds on population. The framework can be extended to deliver bounded population in two natural ways: (i) a hard carrying capacity above which fertility returns to replacement, modeled as a constraint on ψ ; or (ii) endogenous congestion through which the effective time cost rises with population density. Under either extension, $\phi_t \rightarrow 1$ almost surely, but population converges to a stochastic neighborhood of the carrying capacity rather than diverging. The informational poverty trap is unaffected because it operates at populations close to the reference $\bar{N}$, far from any carrying capacity.

Theorem 1 guarantees almost-sure convergence but is silent on the rate. The next analysis shows that convergence can be arbitrarily slow when the economy starts in the trap region.

The Uninformative Region.

Definition 2 (Uninformative Region). *For $\delta > 0$, the δ -uninformative region is $\{N_t : |\ln(N_t/\bar{N})| < \delta\}$. In this region, the per-period information bound is $\mathcal{I}_t < (\theta_R - \theta_M)^2 \delta^2 / (2\sigma_\eta^2) \equiv \bar{\mathcal{I}}(\delta)$.*

When δ is small, $\bar{\mathcal{I}}(\delta)$ is small, and signals reveal almost nothing about the true θ^* .

Definition 3 (Trap Region). *The informational poverty trap region $\mathcal{T}(\delta, \zeta)$ is the set of states (ϕ, N, π) with: (i) $|\ln(N/\bar{N})| < \delta$; (ii) $\bar{n}(\phi) \in [1, 1 + \zeta]$ for small $\zeta > 0$; and (iii) $\pi \leq \phi$, so posterior pessimism aligns with prevalent Malthusianism.*

The third condition is included to ensure the trap is closed under the joint dynamics of (ϕ_t, N_t, π_t) : once $\pi_t \leq \phi_t$, this inequality propagates inductively (see proof of Proposition 3). The next assumption ensures the trap region is non-empty.

Assumption 5 (Fertility Straddles Replacement). *Malthusian fertility is below replacement and Romerian fertility above: $n_M < 1 < n_R$.*

Under Assumption 5, there is a critical Romerian share $\phi^{trap} = (1 - n_M)/(n_R - n_M)$ at which $\bar{n} = 1$. For ϕ slightly above ϕ^{trap} , aggregate fertility slightly exceeds 1 and population grows slowly.

Proposition 3 (Slow Learning in the Trap). *Consider a Romerian world ($\theta^* = \theta_R > 0$) under Assumption 5. Suppose $(\phi_0, N_0, \pi_0) \in \mathcal{T}(\delta, \zeta)$ for small $\delta, \zeta > 0$. For target $\bar{\phi} > \phi_0$, define the escape time $T^{esc} = \inf\{t : \phi_t > \bar{\phi}\}$. Then:*

- (i) $\mathbb{E}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] \leq C_1 \bar{\mathcal{I}}(\delta)$ with $C_1 = 1/4$;
- (ii) $\phi_{t+1}^{birth} - \phi_t \leq C_2(n_R - n_M)(1 - \phi_t)$ with $C_2 = 1/n_M$;
- (iii) $\mathbb{E}[T^{esc}] \geq (\bar{\phi} - \phi_0)/[\lambda C_2(n_R - n_M) + (1 - \lambda)C_1 \bar{\mathcal{I}}(\delta)]$.

The proof is in Appendix A. As $\delta \rightarrow 0$, the learning contribution vanishes and escape relies on demographic selection, which is itself slow when $n_R - n_M$ or λ is small. Corollary 4 in the appendix shows that the expected escape time diverges as $\Omega(1/\delta^2)$ when learning is the binding channel and can exceed any finite horizon.

Proposition 8 in the appendix characterizes the boundary of the trap. The critical population beyond which learning dominates demographic selection is

$$N^{crit}(\phi)/\bar{N} = \exp\left(\sqrt{\frac{2\sigma_\eta^2 \cdot \lambda C_2(n_R - n_M)(1 - \phi)}{(1 - \lambda)C_1(\theta_R - \theta_M)^2}}\right).$$

High noise σ_η^2 raises N^{crit} , making the trap larger. Strong cultural transmission λ also raises N^{crit} by reducing reliance on learning. Large belief dispersion $(\theta_R - \theta_M)^2$ lowers N^{crit} by making signals more diagnostic when population deviates from $\bar{N}$.

In a Romerian world, then, Malthusian minorities cause prolonged stagnation through a self-reinforcing mechanism. Pessimism suppresses fertility, low fertility keeps population near the reference $\bar{N}$ (the uninformative region), and uninformative signals fail to correct pessimism. Escape eventually occurs but can take arbitrarily long.

5.3 Dynamics in a Malthusian World: Opposing Forces and Non-Monotone Outcomes

The dynamics differ qualitatively when $\theta^* < 0$. Whereas in a Romerian world demographic selection and learning push the Romerian share ϕ_t in the same direction, in a Malthusian world they pull in opposite directions, with the balance between them itself depending on the state. The result of this section makes the opposition rigorous. Demographic selection always pushes ϕ_t upward; learning under truth always pushes π_t (and through cultural

transmission ϕ_t) downward; and the relative strength of the two forces flips with $\ln(N_t/\bar{N})$. Monotone convergence is therefore impossible, and the system inherits the potential for non-monotone dynamics, including the boom-bust cycles familiar from pre-industrial demography that Section 6 discusses.

Proposition 4 (Opposing Forces in Malthusian Worlds). *Suppose $\theta^* = \theta_M < 0$. At any interior state with $\phi_t \in (0, 1)$ and $\pi_t \in (0, 1)$:*

- (i) *Demographic-selection drift is strictly positive and bounded. The contribution of differential fertility to the evolution of ϕ_t satisfies*

$$0 < \phi_t^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t} \leq \frac{n_R - n_M}{4n_M},$$

independently of $\ln(N_t/\bar{N})$.

- (ii) *Bayesian learning drift is strictly negative under truth and grows quadratically in population deviation. The expected change in the posterior under the true law of motion is*

$$\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] = -\pi_t(1 - \pi_t)\mathcal{I}_t + o(\mathcal{I}_t), \quad \mathcal{I}_t = \frac{(\theta_R - \theta_M)^2[\ln(N_t/\bar{N})]^2}{2\sigma_\eta^2}.$$

- (iii) *The joint drift in ϕ_t reverses sign as a function of population deviation. The expected one-period drift*

$$\mathbb{E}^{\theta_M}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] = \lambda(\phi_t^{birth} - \phi_t) + (1 - \lambda)\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t]$$

is strictly positive at $\ln(N_t/\bar{N}) = 0$ and strictly negative whenever $|\ln(N_t/\bar{N})|$ exceeds an explicit threshold $\bar{L}(\phi_t, \pi_t, \lambda)$ given in the proof. In particular, the deterministic skeleton admits no monotone law of motion in ϕ_t .

The proof, in Appendix A, is a direct Bayes log-odds calculation combined with a bounded-selection-drift comparison. The threshold $\bar{L}(\phi_t, \pi_t, \lambda)$ takes the explicit form

$$\bar{L}(\phi_t, \pi_t, \lambda)^2 = \frac{2\sigma_\eta^2 \cdot \lambda \phi_t(1 - \phi_t)(n_R - n_M)}{(1 - \lambda)\pi_t(1 - \pi_t)(\theta_R - \theta_M)^2 \bar{n}_t},$$

which is decreasing in the Romerian-Malthusian fertility gap and increasing in signal noise.

Economic content. Proposition 4 captures the core economic asymmetry between the two worlds. In a Romerian world (Theorem 1), demographic selection and learning both

favor Romerians, and trajectories converge monotonically toward $\phi = 1$ almost surely. In a Malthusian world, the two forces conflict. Demographic selection always pushes upward because optimists outbreed pessimists, learning under truth always pushes downward because outcomes confirm the Malthusian model, and the magnitude of the learning channel scales quadratically with how far population has deviated from the reference. The drift is therefore positive near $\bar{N}$, where signals are uninformative and demographic selection dominates, and negative once population has grown enough that the learning channel overtakes selection. The system cannot settle by drifting monotonically in either direction. Any candidate interior steady state must lie at a population strictly above the reference, where the deterministic component of log-productivity growth vanishes: $\bar{g} + \theta_M \ln(N^{ss}/\bar{N}) = 0$ gives $\ln(N^{ss}/\bar{N}) = -\bar{g}/\theta_M > 0$.

Cycle dynamics. The state-dependent reversal in Proposition 4 naturally produces the cyclical pattern familiar from economic-historical accounts of the pre-industrial era, with three identifiable phases. *Expansion phase:* population is moderate, $|\ln(N_t/\bar{N})|$ is small, learning is weak, demographic selection dominates, and ϕ_t rises. *Crisis phase:* as ϕ_t and population grow, signals become informative; productivity slows because $\theta^* < 0$; the Malthusian model fits much better than the Romerian; π_{t+1} falls sharply, overwhelming demographic selection; and ϕ_{t+1} falls. *Recovery phase:* fertility falls, population stabilizes or contracts, signals weaken, demographic selection regains the upper hand, and the pattern can repeat. Whether the system traces out a deterministic limit cycle, damped oscillations around the interior steady state, or stochastic recurrence around it depends on the curvature of the non-linear dynamics and the size of the structural shocks. What Proposition 4 establishes is that all of these belong to the family of non-monotone dynamics that the system can produce, and that none can collapse to monotone convergence.

Comparative statics for the strength of cycles. The threshold $\bar{L}$ and the underlying drift magnitudes deliver clean comparative statics for how pronounced the non-monotone dynamics are. When learning is very weak (very high σ_η^2 or very small belief gap $\theta_R - \theta_M$), $\bar{L}$ is large and the drift remains positive over a wide population range; demographic selection dominates and ϕ_t rises persistently, so the economy lingers in an inefficient high-population, optimist-dominated configuration with low welfare. When learning is very strong (very low σ_η^2), $\bar{L}$ is small and the drift turns negative for modest population deviations; Malthusian beliefs spread quickly, ϕ_t falls toward zero, and the system settles into a Malthusian-dominated regime. The most pronounced non-monotone dynamics, the boom-bust cycles with both phases visible, arise for intermediate learning responsiveness, where neither force dominates

globally and the drift sign reversal repeatedly engages.

5.4 The Asymmetry: Why Wrong Beliefs Generate Different Pathologies

Table 5 summarizes the asymmetry.

Table 5. Pathologies from Wrong Beliefs

	Romerian World ($\theta^* > 0$)	Malthusian World ($\theta^* < 0$)
Wrong belief	Malthusian ($\theta_M < 0$)	Romerian ($\theta_R > 0$)
Demographic selection	Favors Romerians (correct)	Favors Romerians (incorrect)
Learning	Favors Romerians (correct)	Favors Malthusians (correct)
Forces align?	Yes	No
Pathology	Informational poverty trap	Non-monotone dynamics, including cycles
Long-run outcome	Convergence to truth	State-dependent reversal

In Romerian worlds, both forces align toward correct beliefs. Incorrect (Malthusian) beliefs are self-correcting, but the correction can be very slow if the economy starts in the trap. In Malthusian worlds, the forces conflict: demographic selection favors optimists who happen to be wrong, while learning favors the correct pessimistic model. Proposition 4 shows that this conflict produces a state-dependent drift that cannot be resolved by monotone convergence, opening the door to the non-monotone dynamics that economic historians have documented.

This asymmetry has a profound implication. Human societies are more vulnerable to excessive optimism in terms of stability, but more vulnerable to excessive pessimism in terms of stagnation. Optimists reproduce more regardless of whether their optimism is warranted, which destabilizes Malthusian worlds. Pessimism keeps population near the reference and thereby suppresses learning, which traps Romerian worlds.

Remark 2 (The Core Insight). *The central theoretical contribution can be stated sharply: in a Romerian world, Malthusians cause stagnation; in a Malthusian world, Romerians generate non-monotone dynamics, including cycles. Wrong beliefs are always costly, but the nature of the cost depends on the direction of the error.*

6 Demographic History: Two Interpretations

The model provides a unified framework for interpreting broad patterns in demographic history. It admits two fundamentally different interpretations of the pre-industrial era, with distinct implications for understanding the modern world.

6.1 Interpretation 1: A Genuinely Malthusian Past

The conventional interpretation, consistent with economic historians from Malthus himself to [Clark \(2007\)](#), is that the pre-industrial world was genuinely Malthusian: scale effects were negative, and larger populations did reduce living standards.

Under this interpretation, the cycles documented by [Voigtländer and Voth \(2013\)](#) and others (periods of demographic expansion followed by crises that reduced population) reflect exactly the kind of non-monotone dynamics established in Proposition 4. During periods of relative prosperity, Romerian-like optimism spread through demographic selection. Population grew. Eventually, population deviation from the reference became large enough that the learning channel engaged: living standards fell, Malthusian beliefs were reinforced by crisis, and the drift in ϕ_t reversed. Population stabilized or declined, recovery began, and the pattern restarted.

The Black Death of 1347–1351 provides a particularly clear example. Europe’s population had grown substantially in the preceding centuries, straining resources. The plague killed roughly one-third of the population. Real wages rose sharply in the aftermath and remained elevated for over a century. The model interprets this as a crisis phase that both directly reduced population and reinforced Malthusian beliefs, leading to a sustained period of lower fertility.

Under this interpretation, the Industrial Revolution marked a genuine regime change from $\theta^* < 0$ to $\theta^* > 0$. Sustained technological progress, enabled by the accumulation of scientific knowledge and the development of institutions supporting innovation, broke the Malthusian link between population and living standards. The world shifted from being truly Malthusian to truly Romerian.

6.2 Interpretation 2: An Always-Romerian World Trapped by Information Poverty

The informational poverty trap opens a radically different interpretation: the world may have *always* been Romerian, but humanity was trapped for millennia by the self-reinforcing dynamics of pessimism and a population stuck near the reference level.

Under this interpretation, positive scale effects existed throughout history. More people always meant more ideas, more specialization, more innovation. But populations were too close to the relevant reference, and productivity signals too noisy, for anyone to reliably learn this fact. Malthusian beliefs, once established, suppressed fertility, keeping population near the reference $\bar{N}$. A population in this region generated uninformative signals. Uninformative signals failed to correct Malthusian beliefs. The trap persisted not because Malthusianism

was correct, but because the evidence needed to refute it could only be generated by escaping it.

The observed pre-industrial cycles, under this interpretation, were not the inevitable boom-bust dynamics of a Malthusian world but rather failed escape attempts from the trap. Occasionally, some region or era experienced conditions that pushed population higher: better weather, new crops, reduced warfare. Beliefs began to update toward optimism. But the gains were fragile. Any negative shock (plague, famine, invasion) was interpreted as confirmation of Malthusian fears, beliefs snapped back, fertility fell, and the economy returned to the trap.

The Industrial Revolution, under this interpretation, was not a technological regime change but an *informational escape*. Some combination of factors (the Enlightenment, improving institutions, a sequence of positive productivity shocks) pushed European populations above the critical threshold N^{crit} . Once in the informative region, learning accelerated. Evidence of positive scale effects accumulated. Beliefs shifted decisively toward Romerianism. Fertility rose further. Population exploded. The trap dissolved, not because the world changed, but because humanity finally accumulated enough information to see what had always been true.

6.3 Distinguishing the Interpretations

Both interpretations are consistent with the qualitative features of demographic history: long periods of stagnation, occasional expansions, crises, and eventual escape to modern growth. Distinguishing them requires careful attention to details of the historical record and to quantitative magnitudes.

Several features favor the genuine-regime-change interpretation. First, the timing of the Industrial Revolution coincides with specific technological innovations (steam power, mechanization) that plausibly altered the relationship between population and productivity. Second, the scale of technological progress after 1800 suggests a qualitative break rather than just informational updating. Third, environmental constraints (land, resources) appear to have genuinely bound pre-industrial economies in ways that seem inconsistent with large positive scale effects.

Several features favor the informational-escape interpretation. First, there is evidence of positive scale effects even in pre-industrial settings: larger cities were more innovative, and more connected regions were more prosperous. Second, the speed of the transition, once it began, is more consistent with belief updating than with gradual technological change. Third, the modern fertility decline in developed countries (discussed below) is puzzling if the world is truly Romerian, but explicable if beliefs can drift back toward Malthusianism even

in a Romerian world.

I do not take a definitive stand on which interpretation is correct. The contribution of the model is to show that *both are theoretically coherent* and that they have *different implications for policy*.

6.4 The Modern Fertility Decline

Fertility rates in developed countries have fallen dramatically since the 1960s, reaching levels well below replacement in most of Europe and East Asia. This decline is puzzling from a simple Romerian perspective: if larger populations are beneficial, why are the most prosperous societies choosing the fewest children? The model suggests several interpretations.

Revival of Malthusian beliefs. The environmental movement, which emerged prominently in the 1960s and 1970s, can be understood as a revival of Malthusian thinking. Concerns about pollution, resource depletion, climate change, and ecological limits represent beliefs that population growth is harmful. The empirical evidence in Section 3 documents that such beliefs are prevalent and are associated with substantially lower fertility. Under the informational-escape interpretation, the modern fertility decline represents a partial return toward the trap.

Actual shift to Malthusian technology. Under the regime-change interpretation, the true θ^* may have recently shifted from positive to negative. Automation and AI may have reduced the marginal contribution of human labor to economic output. Environmental constraints, particularly climate change, may have imposed genuine costs on population growth.

Quantity-quality substitution. The model abstracts from human capital, but in reality, parents face a quantity-quality tradeoff. Rising returns to education may have induced parents to invest more in each child's human capital at the expense of having more children. This is conceptually distinct from Malthusianism but generates similar fertility outcomes.

Under the dynastic welfare criterion of Section 7, what matters is whether households' beliefs match the truth: information policies that correct misperceptions are warranted only if beliefs are wrong. Under the total utilitarian criterion, the planner additionally values population for its own sake, so even a correct-belief decentralized equilibrium leaves room for intervention through fertility subsidies or optimism-promoting communication.

6.5 Contemporary Divergence and Future Dynamics

Fertility today varies enormously across countries and groups. Total fertility rates range from below 1.0 in some East Asian countries to above 6.0 in parts of Sub-Saharan Africa. Within countries, fertility varies systematically with religion, education, and political orientation.

The model predicts that these differences are partially driven by belief heterogeneity and are self-reinforcing through demographic selection. Groups with Romerian beliefs (or with religious or cultural traditions that encourage high fertility for other reasons) will grow in population share over time. If cultural transmission is strong, these compositional shifts can substantially affect aggregate beliefs and fertility over generations. Standard demographic projections typically assume convergence to low fertility as countries develop, but the model suggests convergence may be slower or incomplete if belief heterogeneity persists. Whether this is efficient depends on the true value of θ^* : if Romerian, high-fertility groups move the population toward efficiency; if Malthusian, they exacerbate the problem.

7 Policy: Welfare Criteria and the Value of Public Information

I now analyze the welfare effects of public announcements about the scale-effect parameter θ . A distinctive feature is the careful treatment of welfare evaluation when population is endogenous. As [Goloso et al. \(2007\)](#) emphasize, standard welfare theorems fail when the set of agents depends on allocations. *Throughout this section, I impose Assumption 4: the policy analysis is conducted in the local-uniqueness region of the global game.*

7.1 Welfare Criteria with Endogenous Population

Evaluating welfare when fertility is a choice variable raises fundamental conceptual issues. The set of people who exist depends on policy, so we cannot simply ask whether everyone is better off; the “everyone” changes. Following [Goloso et al. \(2007\)](#) and the population ethics literature surveyed by [Blackorby et al. \(2005\)](#), I focus on two normatively distinct criteria. (An earlier draft considered a third, “paternalistic” criterion that evaluated dynastic streams under truth. Under the log Barro-Becker structure with the time-cost specification, this criterion is not algebraically distinct from a population-weighted utilitarian one in the directions that matter for the announcement problem, and I drop it.)

Dynastic Welfare. Under the dynastic criterion, the planner respects households’ subjective preferences:

$$W^{dyn} = \phi_0 V_R(A_0, N_0, \phi_0, \pi_0) + (1 - \phi_0) V_M(A_0, N_0, \phi_0, \pi_0). \quad (21)$$

This criterion is closely related to the P-efficiency concept of [Goloso et al. \(2007\)](#), which compares allocations using the welfare of the agents who exist in the original allocation. The dynastic criterion as written here uses subjective expected utility rather than truth-evaluation, so it is the appropriate analogue when the planner is constrained to respect preferences as stated.

Total Utilitarian Welfare. The total utilitarian planner sums period utilities across all individuals alive at each date, evaluated under truth:

$$W^{tot} = \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t N_t \cdot \ln c_t \right]. \quad (22)$$

For (22) to be well-defined, $\beta \mathbb{E}^* n_R^* < 1$ at the relevant equilibrium fertility. Under the time-cost specification, $\ln c_t = \ln A_t + \ln(1 - \psi(n_t))$, which decomposes additively into a productivity term and a fertility-cost term, making the planner's tradeoffs transparent.

Average utilitarian benchmark. For comparison, define

$$W^{avg} = \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t \ln c_t \right]. \quad (23)$$

A clarification is warranted. W^{avg} does internalize the per-capita scale-effect channel, since $\ln c_t$ depends on $\ln A_t$, which depends on past population through (5). What W^{avg} omits relative to W^{tot} is the *population-multiplier* term N_t , which gives the total utilitarian planner an additional motive to value larger populations beyond the per-capita welfare effect. The Repugnant Conclusion of [Parfit \(1984\)](#) is the strong-form expression of this difference.

Remark 3 (Choice of Welfare Criterion). *The dynastic and total utilitarian criteria span a substantively distinct normative spectrum. The former treats existing households as the moral universe and respects their preferences; the latter treats all individuals who ever exist as the moral universe and evaluates outcomes under truth. The contrast is the locus of the genuinely interesting policy results: the dynastic planner is fundamentally an information provider, while the total utilitarian planner is an externality corrector. The average-utilitarian benchmark sits between them, internalizing the per-capita externality but not the population motive.*

7.2 The First-Best Allocation and Welfare Decomposition

A planner who knows θ^* and can directly choose a symmetric fertility path under the total utilitarian criterion solves

$$\max_{\{n_t\}} \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t N_t \cdot (\ln A_t + \ln(1 - \psi(n_t))) \right] \quad (24)$$

subject to $N_{t+1} = n_t N_t$ and (4). The planner internalizes the effect of fertility on future TFP through the scale effect. Conjecturing a value function of the local log-linear form $V^P(A, N) = N \cdot w^P(A, N)$ with $w^P(A, N) \approx v^P + \xi_A^P \ln A + \xi_N^P \ln(N/\bar{N})$ (with the same local-approximation caveats as in Section 2), the planner's FOC for fertility is

$$\frac{\psi'(n_t^{FB})}{1 - \psi(n_t^{FB})} = \beta [w^P(A', N') + \xi_N^P]. \quad (25)$$

The household's common-correct-belief FOC is structurally identical except for the absence of ξ_N^P . The wedge between planner and household optimality is therefore exactly $\beta \xi_N^P$, with the sign of θ^* .

Proposition 5 (First-Best vs. Decentralized Fertility). *In a Romerian world ($\theta^* > 0$): (i) $n^{FB} > n^{CB}$, where n^{CB} is the common-correct-belief decentralized fertility; (ii) $n^{FB} > n^{CB} > n_M^*$; (iii) the welfare gap decomposes into externality, mean-bias, and dispersion components. In a Malthusian world ($\theta^* < 0$), $n^{FB} < n^{CB}$.*

The proof is in Appendix B. In decentralized equilibrium, households do not internalize the scale-effect externality. In a Romerian world, this leads to under-investment even with correct beliefs, and belief heterogeneity adds two further sources of inefficiency.

Proposition 6 (Welfare Decomposition). *Under the total utilitarian criterion,*

$$W^{FB}(\theta^*) - W^{HB} = \underbrace{\Delta_1(\theta^*)}_{\text{externality}} + \underbrace{\Delta_2(\theta^*, \bar{\theta})}_{\text{mean bias}} + \underbrace{\Delta_3(\text{Var}(\theta_i))}_{\text{dispersion}}, \quad (26)$$

where $\Delta_1 \geq 0$ with equality iff $\theta^* = 0$; $\Delta_2 \geq 0$ increasing in $(\bar{\theta} - \theta^*)^2$; and $\Delta_3 \approx (\omega \kappa^2 / 2) \text{Var}(\theta_i)$. The Taylor expansion underlying Δ_3 is taken around the social-optimum fertility n^{SP} , so the externality Δ_1 appears explicitly rather than being absorbed into the dispersion remainder.

The proof is in Appendix B. Public announcements can reduce Δ_2 (shifting mean beliefs toward truth) and Δ_3 (reducing dispersion) but cannot directly address Δ_1 without additional policy instruments such as fertility subsidies. A planner with both an information

instrument and a fertility instrument could close all three gaps. A planner with only the information instrument can close Δ_2 and Δ_3 directly and, as I show next, use the announcement strategically to compensate *partially* for Δ_1 .

Interpretation. The decomposition has three economically distinct messages. First, even with average beliefs perfectly calibrated ($\bar{\theta} = \theta^*$, so $\Delta_2 = 0$), the externality Δ_1 persists: households do not internalize the productivity spillover from their own fertility. Second, even with no dispersion and correct mean beliefs, the externality remains. Third, dispersion generates welfare loss even when $\bar{\theta} = \theta^*$, because households' fertility choices cease to coincide with the symmetric social optimum.

7.3 The Planner's Announcement Problem

Consider a planner who observes a noisy estimate of the true parameter:

$$\hat{\theta} = \theta^* + \nu, \quad \nu \sim \mathcal{N}(0, \sigma_\nu^2). \quad (27)$$

The planner publicly releases a signal $z = \mu + \tilde{\nu}$, where μ is the chosen announcement value and $\tilde{\nu} \sim \mathcal{N}(0, \tau_A^{-1})$ is the (exogenous) noise households perceive. Households use z in their Bayes update over the two states $\{\theta_R, \theta_M\}$, updating

$$\pi_t^{post} = \frac{\pi_t \cdot \tilde{\ell}(z \mid \theta_R)}{\pi_t \cdot \tilde{\ell}(z \mid \theta_R) + (1 - \pi_t) \cdot \tilde{\ell}(z \mid \theta_M)}, \quad (28)$$

where $\tilde{\ell}(z \mid \theta) = (\tau_A/2\pi)^{1/2} \exp(-\tau_A(z - \theta)^2/2)$. For the comparative statics that follow, the relevant summary of the announcement is its implied posterior mean $\theta_{post}^A \equiv \mathbb{E}[\theta \mid z, \pi_t]$, which is a smooth function of μ , π_t , and τ_A . With slight abuse of notation, I refer to μ as the announcement and characterize the planner's optimal choice through its effect on θ_{post}^A . For households of intermediate priors and announcement values, the linearization

$$\theta_j^{post} = \frac{\tau_j \theta_j + \tau_A \mu}{\tau_j + \tau_A} \quad (29)$$

provides a tractable approximation that I use in the local comparative statics. The formal result holds under (28), and (29) should be read as a reduced-form summary of the binary update.

Assumption 6 (Announcement Credibility). *Households treat the announcement as a Gaussian signal with precision $\tau_A > 0$. When $\tau_A \rightarrow \infty$, households fully adopt the announced value; when $\tau_A \rightarrow 0$, households ignore the announcement.*

I take τ_A as exogenous. A microfoundation through repeated-game reputation would make τ_A the equilibrium of a sub-game in which households update on the planner’s history of announcements, and would constrain optimal shading by binding it to future credibility. I treat τ_A as a comparative static and flag a full microfoundation as an extension.

7.4 Optimal Announcement under Different Criteria

Theorem 2 (Optimal Announcement: Dynastic Welfare). *Suppose Assumption 4 holds. Under W^{dyn} , the optimal announcement is truthful in expectation: $\mu^{dyn} = \hat{\theta}$, and the planner prefers maximal informativeness.*

Sketch. Under subjective preferences, an information signal is welfare-increasing in the Blackwell sense whenever it reveals more about θ^* . Among all signals with prior mean $\hat{\theta}$, a deterministic signal at $\mu = \hat{\theta}$ maximally reveals the planner’s posterior, and any deterministic deviation makes the signal strictly less informative. The full proof in Appendix B verifies that this remains optimal among stochastic announcement rules satisfying the Gaussian likelihood structure of (28). $\square$

The economic content is sharp. Under the dynastic criterion, the planner is fundamentally an information provider. There is no externality the planner cares about beyond what households themselves would care about, so distorting the announcement strictly reduces welfare.

Theorem 3 (Optimal Announcement: Total Utilitarian Welfare). *Suppose Assumption 4 holds, and consider $\theta^* = \theta_R > 0$. Under W^{tot} , the optimal announcement satisfies*

$$\mu^{tot} = \hat{\theta} + \chi^{tot}, \quad (30)$$

where $\chi^{tot} > 0$ is an “optimism shade” whenever the externality $\Delta_1 > 0$. The shade has the following comparative statics: (i) χ^{tot} is increasing in $|\theta^*|$; (ii) χ^{tot} is increasing in $|\ln(N_0/\bar{N})|$; (iii) χ^{tot} is decreasing in $\text{Var}(\theta_{i,0})$; (iv) χ^{tot} is increasing in announcement credibility τ_A . In a Malthusian world ($\theta^* < 0$), the sign of χ^{tot} reverses: the planner shades toward pessimism.

The proof is in Appendix B. The intuition is that the planner equates the marginal utilitarian benefit of shifting average fertility (carrying an externality wedge of size $\beta\xi_N^P$) to the marginal cost of distorting individual fertility decisions away from their belief-conditional optimum.

Corollary 1 (Optimal Announcement: Average Utilitarian Welfare). *Under W^{avg} , the optimal announcement is $\mu^{avg} = \hat{\theta} + \chi^{avg}$, where $0 < \chi^{avg} < \chi^{tot}$ in a Romerian world (with the inequalities reversing in a Malthusian world). The wedge driving χ^{avg} is the per-capita scale-effect spillover $\beta\xi_{N,avg}^P$, which is strictly smaller in absolute value than the total utilitarian wedge $\beta\xi_N^P$ because it omits the population-multiplier term N_t from (22).*

The proof is in Appendix B. The corollary closes the welfare-criterion-to-shading map. The dynastic planner does not shade at all; the average-utilitarian planner shades to internalize the per-capita externality; the total utilitarian planner shades further to also internalize the population motive.

Corollary 2 (Ordering of Announcements). *Under Assumption 4 and in a Romerian world, $\mu^{dyn} = \hat{\theta} < \mu^{avg} < \mu^{tot}$, with the inequalities reversing in a Malthusian world. The total utilitarian planner promotes Romerian beliefs more aggressively than the average utilitarian, who in turn promotes them more aggressively than the dynastic planner.*

The ordering captures the substantive contrast across criteria. Respecting subjective preferences (dynastic), internalizing only the per-capita externality (average utilitarian), or additionally valuing population (total utilitarian) imply systematically different communication strategies, with shading magnitudes that strictly increase as the planner’s normative commitments expand.

7.5 The Planner’s Dilemma and the State-Dependent Value of Information

A key insight emerges from the structure of the learning problem: the planner faces the same informational constraints as households. If the economy is in the uninformative region, the historical productivity signals that inform $\hat{\theta}$ are themselves uninformative. The planner’s signal precision is endogenously low precisely when the trap is most binding.

Remark 4 (The Planner’s Dilemma). *Public information is most valuable in the informational poverty trap, where coordinating beliefs could push population above N^{crit} and initiate escape. But this is precisely when the planner has the least information. The cases where intervention matters most are the cases where the planner is most uncertain about what to say.*

This creates a policy dilemma distinct from Morris and Shin (2002). There, the concern is that public information may be *too influential*; here, the planner may have *nothing useful to say* when saying something would matter most.

Trap-region value. Appendix B, Proposition 9, shows that in the trap region in a Roman world, the expected value of truthful announcement under W^{tot} is $\mathcal{V}^{trap} = P^{esc} \cdot \Delta W^{esc} - P^{deepen} \cdot \Delta W^{deepen}$, with a precision threshold above which $\mathcal{V}^{trap} > 0$. Because ΔW^{esc} is large (escape avoids extended stagnation), even a moderately noisy signal can be worth announcing.

Non-monotone-region value. Proposition 10 establishes that in a Malthusian world, where the dynamics are non-monotone by Proposition 4, the value of announcement comes from reducing belief dispersion (and thereby dampening oscillations and accelerating convergence toward the interior steady state). This benefit is partially offset by adjustment costs from disrupting the existing dynamics. For sufficiently precise signals the value is positive, but its magnitude is smaller than the trap-escape gain.

The stabilizing role of Malthusians. An additional nuance distinguishes my setting from pure-complementarity coordination games. In response to a public announcement, Romanians coordinate (amplifying the signal), but Malthusians anti-coordinate (dampening the signal). This heterogeneity reduces the risk of over-coordination on noise. Compared to a population of only Romanians, a mixed population is safer to inform. The precision threshold for beneficial announcement is therefore *lower* in populations with substantial Malthusian minorities. Paradoxically, the presence of pessimists makes communication policy safer.

7.6 Comparison of Decentralized and Planner Allocations

Table 6 summarizes the contrast across criteria.

Table 6. Decentralized vs. Planner Allocations by Welfare Criterion

	Dynastic	Average Util.	Total Util.
<i>Romeroian World</i> ($\theta^* > 0$)			
Decentralized fertility		Too low (externality + wrong beliefs)	
Optimal announcement	Truthful	Mildly shaded toward θ_R	Strongly shaded toward θ_R
Residual inefficiency	Externality only	Reduced externality	Most reduced externality
<i>Malthusian World</i> ($\theta^* < 0$)			
Decentralized fertility		Too high (externality), mixed by beliefs	
Optimal announcement	Truthful	Mildly shaded toward θ_M	Strongly shaded toward θ_M
Residual inefficiency	Externality only	Reduced externality	Most reduced externality

Under all three criteria, the decentralized equilibrium is inefficient due to the uninternalized scale-effect externality. The dynastic planner can address belief distortions but not the

externality. The average-utilitarian planner additionally uses the announcement as a partial substitute for an unavailable per-capita-fertility instrument. The total utilitarian planner shades further still, reflecting an additional motive to value population. Additional instruments (fertility subsidies in Romerian worlds, fertility taxes in Malthusian worlds) would be needed to achieve full efficiency under any of the criteria.

7.7 Policy Implications

The analysis yields several implications for government communication policy.

In the trap, take calculated risks. When the economy appears trapped (low fertility, stagnant population, persistent pessimism), even a moderately imprecise signal may be worth announcing if it has a chance of coordinating beliefs and initiating escape. Under the total utilitarian criterion, the expected value of escape justifies risk-taking.

Under dynastic welfare, be truthful. If the planner respects household preferences as given, the optimal policy is to provide accurate information. Distorting announcements would make some households worse off according to their own preferences, and the planner has nothing else to gain.

Under utilitarian criteria, shade systematically. The total utilitarian planner shades announcements toward optimism in Romerian worlds and toward pessimism in Malthusian worlds, by an amount that scales with the externality, the deviation of population from the reference, and credibility. The average-utilitarian planner shades less but in the same direction. Either form of shading is a substitute, not a complement, for fertility instruments.

During Malthusian non-monotone dynamics, time announcements carefully. In a Malthusian world, crises naturally correct beliefs through the learning channel of Proposition 4. Announcing during an expansion (when population is high and signals are informative) can accelerate learning and dampen the swings. Announcing during a crisis is less valuable because beliefs are already moving toward Malthusianism.

Build credibility for when it matters; recognize the limits of knowledge. The value of announcement depends on credibility. Planners should invest in credibility during normal times (by making accurate predictions) so that announcements are effective when intervention matters most. The most important policy implication, however, is epistemic humility. When the economy is in the trap, the planner knows the least about what to say.

Acknowledging this uncertainty, rather than projecting false confidence, may be the most honest policy.

8 Conclusion

Beliefs about the economic consequences of population are not merely background assumptions in fertility models. They are first-order determinants of demographic and macroeconomic dynamics. This paper has formalized that observation in an overlapping generations model with heterogeneous beliefs about scale effects and a time cost of children, with multiplicative log-normal productivity dynamics that admit a tractable first-order log-linear approximation of the value function around a balanced-growth reference. The resulting feedback between beliefs, fertility, and the informativeness of data produces a sharp duality. Malthusian minorities trap Romerian economies in stagnation, while Romerian minorities generate non-monotone dynamics in Malthusian economies, ruling out the monotone convergence available when the truth is Romerian and opening the door to the boom-bust patterns that economic historians have long documented. Wrong beliefs are always costly, but the form of the cost is determined by the truth they fail to track.

The informational poverty trap is the paper's most distinctive theoretical object. Unlike traps grounded in technological nonconvexities or credit frictions, it arises endogenously from the geometry of Bayesian learning. Signals are uninformative precisely in the region where pessimism keeps the economy confined, so the very beliefs that cause stagnation also insulate themselves from correction. This mechanism reframes a long-standing question in economic history. The pre-industrial Malthusian era can be read as a true Malthusian regime that ended with a technological break, or as a Romerian regime trapped for millennia by self-reinforcing pessimism that finally dissolved when population breached the informative region. The two readings are observationally similar in qualitative terms but carry sharply different implications for whether the modern fertility decline represents a benign demographic adjustment or a slow drift back into the trap.

Belief heterogeneity also forces a reckoning with how welfare itself is defined. Because the population over which welfare is evaluated is endogenous to the policy being evaluated, the dynastic, average-utilitarian, and total utilitarian criteria recommend systematically different communication strategies, with shading magnitudes that strictly increase as the planner's normative commitments expand. The dynastic planner is fundamentally an information provider; the average-utilitarian planner internalizes only the per-capita externality; the total utilitarian planner additionally values population. The same logic produces an unconventional message about public information: the binding constraint is that the planner

is most uncertain exactly when announcements would be most valuable. The heterogeneous strategic interaction softens but does not resolve this tension, and is what permits the policy analysis to be conducted within the local-uniqueness region defined by the contraction condition on the aggregate best response.

Several extensions are natural. A complete classification of the deterministic skeleton in Malthusian worlds, going beyond the non-monotonicity established here to characterize when the system produces damped oscillations, stochastic recurrence, or sustained limit cycles, would deepen the connection between the theory and the historical record on pre-industrial demographic cycles. A custom global-games selection theorem suited to the heterogeneous-strategic-interaction setting would extend the policy analysis beyond the local-uniqueness region. The empirical results establish a robust conditional correlation but stop short of causal identification; instrumental variation in belief exposure, through plausibly exogenous shifts in environmental discourse or schooling content, would sharpen the case. The model abstracts from the quantity-quality margin central to unified growth theory, and embedding human capital choice would let the framework speak directly to the demographic transition. Open-economy extensions with migration would introduce belief-driven sorting across borders. A microfoundation for announcement credibility through repeated-game reputation, and a fully bounded-population specification through carrying capacity or congestion, would extend the analysis. Finally, a quantitative implementation disciplined by long-run population and productivity data could in principle adjudicate between the two interpretations of pre-industrial history that the theory leaves open.

These extensions are worth pursuing because the framework joins endogenous fertility (which typically takes the population-productivity relationship as known) with heterogeneous beliefs (which typically take the set of agents as fixed). The informational poverty trap, the Romerian-Malthusian duality, and the state-dependent value of public information all follow from taking the joint evolution of population and beliefs seriously. Whether one reads human history as a Malthusian regime that ended or a Romerian regime that nearly never began, the lesson is the same. What societies believe about themselves shapes what they become, and what they become determines what they can learn.

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A Proofs and Auxiliary Theoretical Results

A note on the appendix proofs

Several appendix results below rely on the local log-linear approximation (16) of the household value function. Throughout the appendix I distinguish between (a) parts of the argument that are exact under the multiplicative log-normal technology of (4), and (b) parts that hold to first order around a balanced-growth reference. The proof of Proposition 4 is a direct calculation that does not rely on a dynamical-systems reduction; the global-games selection sketch in Proposition 7 is explicitly heuristic and is not used in any policy result.

A.1 Proof of Lemma 1

Step 1: Bellman equation. Substituting $c_{j,t} = A_t(1 - \psi(n))$ and $\ln c_{j,t} = \ln A_t + \ln(1 - \psi(n))$ into the dynastic objective gives (15). Atomistic households take aggregate fertility $\bar{n}$ and the laws of motion for (A', N', ϕ', π') as given; the choice variable n enters the period return only through $\ln(1 - \psi(n))$ and the discount-multiplier term βn .

Step 2: Local log-linear conjecture. Conjecture

$$V_j(A, N, \phi, \pi) \approx v_j(\phi, \pi) + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}), \quad (31)$$

to be interpreted as a first-order approximation around a balanced-growth reference at which n_j^* is locally constant in $(\ln A, \ln(N/\bar{N}))$. The constants $\xi_{A,j}, \xi_{N,j}$ below are the values of the corresponding partial derivatives of V_j at the reference point.

Step 3: Expected continuation value. Under the multiplicative log-normal TFP (4),

$$\ln A' = \ln A + \bar{g} + \theta^* \ln(N/\bar{N}) + \eta_{t+1}$$

exactly. Taking expectations under belief θ_j , with $\mathbb{E}_j[\eta_{t+1}] = 0$:

$$\mathbb{E}_j[\ln A'] = \ln A + \bar{g} + \theta_j \ln(N/\bar{N}).$$

For population, $\ln(N'/\bar{N}) = \ln(N/\bar{N}) + \ln \bar{n}$. Substituting (31) forward:

$$\mathbb{E}_j[V_j(A', N', \phi', \pi')] \approx \mathbb{E}_j v_j(\phi', \pi') + \xi_{A,j} [\ln A + \bar{g} + \theta_j \ln(N/\bar{N})] + \xi_{N,j} [\ln(N/\bar{N}) + \ln \bar{n}]. \quad (32)$$

The technology side of the Bellman contributes no Taylor remainder; the only approximation enters through the conjecture (31).

Step 4: Coefficient matching at the reference point. Substituting (32) into the Bellman and evaluating at the optimum n_j^* :

$$\begin{aligned} v_j(\phi, \pi) + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}) &= \ln A + \ln(1 - \psi(n_j^*)) \\ &+ \beta n_j^* \left[\mathbb{E}_j v_j(\phi', \pi') + \xi_{A,j} (\ln A + \bar{g} + \theta_j \ln(N/\bar{N})) \right. \\ &\quad \left. + \xi_{N,j} (\ln(N/\bar{N}) + \ln \bar{n}) \right]. \end{aligned}$$

Treating n_j^* as locally constant at the reference point, matching coefficients on $\ln A$:

$$\xi_{A,j} = 1 + \beta n_j^* \xi_{A,j} \implies \xi_{A,j} = \frac{1}{1 - \beta n_j^*}.$$

Matching coefficients on $\ln(N/\bar{N})$:

$$\xi_{N,j} = \beta n_j^* (\xi_{A,j} \theta_j + \xi_{N,j}) \implies \xi_{N,j} = \frac{\beta n_j^* \theta_j}{(1 - \beta n_j^*)^2}.$$

Matching constants:

$$v_j(\phi, \pi) = \ln(1 - \psi(n_j^*)) + \beta n_j^* [\mathbb{E}_j v_j(\phi', \pi') + \xi_{A,j} \bar{g} + \xi_{N,j} \ln \bar{n}].$$

Step 5: Status of the conjecture. The matching above treats n_j^* as constant in $(\ln A, \ln(N/\bar{N}))$.

Step 6 below shows via the FOC that n_j^* inherits state dependence through $\xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N})$ on the right-hand side of the optimality condition. *Two observations follow.* First, the multiplicative log-normal specification eliminates the technology-side Taylor remainder, so equation (32) is exact in $(\ln A, \ln(N/\bar{N}))$ conditional on the local-constancy of n_j^* . Second, the resulting conjecture is therefore a first-order log-linear approximation around the balanced-growth reference, and the closed-form coefficients reported above are local objects. The qualitative results elsewhere in the paper rely only on the sign of $\xi_{N,j}$, which inherits the sign of θ_j under any value function with smooth dependence on the state.

Step 6: First-order condition. Differentiating the right-hand side of the Bellman with respect to n :

$$-\frac{\psi'(n_j^*)}{1 - \psi(n_j^*)} + \beta \mathbb{E}_j [V_j(A', N', \phi', \pi')] = 0,$$

which is (17). Strict monotonicity of the first term in n_j^* follows from

$$\frac{d}{dn} \left[\frac{\psi'(n)}{1 - \psi(n)} \right] = \frac{\psi''(n)(1 - \psi(n)) + (\psi'(n))^2}{(1 - \psi(n))^2} > 0$$

under $\psi' > 0, \psi'' \geq 0$. Under the parametric form (6), $\psi'(n)/(1 - \psi(n)) = \gamma n$, so the FOC reduces to

$$\gamma n_j^* = \beta \mathbb{E}_j[V_j(A', N', \phi', \pi')].$$

Step 7: Fixed point and IFT. The system formed by Step 4 (which expresses $\xi_{A,j}, \xi_{N,j}, v_j$ in terms of n_j^* , at the reference point) and Step 6 (which expresses n_j^* in terms of the continuation value) is a fixed-point problem in n_j^* . Under $\beta n_j^* < 1$, the Jacobian is non-singular: the LHS of the reduced FOC has slope $\gamma > 0$ in n_j^* , while the RHS has slope of order $\beta/(1 - \beta n_j^*)^2$, which is finite and positive. The implicit function theorem applies and yields a unique n_j^* at the reference. Differentiating the system with respect to θ_j shows that n_j^* is strictly increasing in θ_j . Writing $n_j^* = n^* h(\theta_j; \bar{n})$ for a baseline n^* and an increasing function h gives the form claimed in the lemma. Since $\theta_R > \theta_M$, $n_R^* > n_M^*$. $\square$

Remark on type switching. The Bellman treats descendants as type- j with probability one. As discussed in Remark 1, an exact formulation would replace V_j on the right-hand side with $\lambda V_j + (1 - \lambda)V_{\text{learn}}$. The coefficient algebra in Step 4 generalizes with weights $\lambda + (1 - \lambda)q_j$ replacing 1, where q_j is the parent's subjective probability that a learning child adopts type j ; the qualitative ranking $n_R^* > n_M^*$ is preserved.

A.2 Proof of Proposition 1

Step 1: Setup. For type j expecting $\bar{n}^e$,

$$\mathbb{E}_j[V_j] \approx v_j(\phi', \pi') + \xi_{A,j}[\ln A + \bar{g} + \theta_j \ln(N/\bar{N})] + \xi_{N,j}[\ln(N/\bar{N}) + \ln \bar{n}^e].$$

The dependence of $v_j(\phi', \pi')$ on $\bar{n}^e$ (through ϕ', π') is of higher order; the dominant direct dependence is through $\xi_{N,j} \ln \bar{n}^e$.

Step 2: Sign of marginal continuation value.

$$\frac{\partial \mathbb{E}_j[V_j]}{\partial \bar{n}^e} = \frac{\xi_{N,j}}{\bar{n}^e}.$$

At the local approximation, $\xi_{N,j}$ has the sign of θ_j by (18): $\xi_{N,M} < 0$ and $\xi_{N,R} > 0$. The sign claim is robust beyond the approximation: under any smooth value function, $\partial V_j / \partial \ln N$ inherits the sign of θ_j at any state where the marginal contribution of population to expected log-productivity is well-defined.

Step 3: Effect on optimal fertility. The FOC has a strictly increasing LHS in n_j , so by IFT, $\text{sign}(dn_j^*/d\bar{n}^e) = \text{sign}(\xi_{N,j}) = \text{sign}(\theta_j)$. $\square$

A.3 Slope of the Aggregate Best Response (Auxiliary Result)

Corollary 3. *Under Proposition 1, the aggregate best response $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1 - \phi)n_M^*(\bar{n}^e)$ has slope $\rho(\phi)$ defined in (19): positive for large ϕ , negative for small ϕ , zero at ϕ^* (evaluated locally at the relevant $\bar{n}^e$).*

Proof. Direct from Proposition 1 and continuity in ϕ . $\square$

A.4 Proof of Proposition 2

Step 1. Equilibrium requires $\bar{n}^e = BR(\bar{n}^e)$. *Step 2.* The contraction-mapping argument shows that BR admits a locally unique fixed point in a neighborhood of the relevant $\bar{n}^e$ when $|\rho(\phi)| < 1$; tangency to the 45-degree line at $|\rho(\phi)| = 1$; when $\rho(\phi) > 1$, the slope locally exceeds the 45-degree line and BR can intersect it transversally at multiple points, generating multiple locally ordered equilibria. *Step 3.* The condition is local and is evaluated at the candidate fixed point. $\square$

A.5 Heuristic Equilibrium Selection (Auxiliary Result)

Proposition 7 (Heuristic Selection). *Consider the parameter region where Proposition 2(ii) holds. Suppose households observe private signals $\phi_i = \phi + \zeta_i$. As $\sigma_\zeta \rightarrow 0$, a heuristic threshold-equilibrium argument selects a candidate equilibrium with a common cutoff ϕ^{**} : Romerians coordinate on the high-fertility branch when $\phi_i > \phi^{**}$; Malthusians, by strategic substitutability, choose their low-fertility response in the same region. The argument is explicitly heuristic. A fully rigorous selection theorem in this heterogeneous-strategic-interaction setting would require an analog of Frankel et al. (2003), which is not available here, and this paper does not rely on the selection in any of its policy results.*

Proof sketch. Adapt Morris and Shin (1998). At ϕ^{**} , each type must be indifferent. As $\sigma_\zeta \rightarrow 0$, the rank distribution of ϕ^{**} becomes uniform. Romerian and Malthusian indifference

functions have opposite slopes; their intersection yields a unique ϕ^{**} under standard regularity. The non-rigorous step is the global characterization of best responses given threshold beliefs: in heterogeneous-strategic-interaction games with continuous actions, the standard iterated-deletion argument used by [Morris and Shin \(1998\)](#) requires additional structure that is not imposed here. $\square$

A.6 Proof of Theorem 1

Step 1: Demographic selection. For interior ϕ_t , $\phi_t^{birth} - \phi_t = \phi_t(1 - \phi_t)(n_R - n_M)/\bar{n}_t > 0$ strictly.

Step 2: Posterior dynamics. Define log-odds $L_t = \ln[(1 - \pi_t)/\pi_t]$. Bayes' rule (12) gives $L_{t+1} - L_t = \ln[\ell(s_t | \theta_M)/\ell(s_t | \theta_R)]$. Using $s_t = \bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}$ and taking expectations under truth $\theta^* = \theta_R$:

$$\mathbb{E}^{\theta_R}[L_{t+1} - L_t | \mathcal{F}_t] = -\frac{(\theta_R - \theta_M)^2 [\ln(N_t/\bar{N})]^2}{2\sigma_\eta^2} = -\mathcal{I}_t \leq 0.$$

By Bayesian consistency, if $\sum_t \mathcal{I}_t = \infty$ a.s., then $L_t \rightarrow -\infty$ and $\pi_t \rightarrow 1$ a.s.

Step 3: Cumulative information diverges. Suppose toward contradiction that $\bar{n}_t \rightarrow 1$ along a subsequence of full asymptotic density. Step 1 gives a deterministic positive drift in ϕ_t^{birth} , so ϕ_t eventually exceeds ϕ^{trap} and $\bar{n}_t > 1 + \zeta$ for some $\zeta > 0$, contradicting $\bar{n}_t \rightarrow 1$. Hence $|\ln(N_t/\bar{N})|$ is bounded away from 0 with positive probability infinitely often, and $\sum \mathcal{I}_t = \infty$ a.s.

Step 4: $\phi_t \rightarrow 1$ a.s. Given $\pi_t \rightarrow 1$, for any $\varepsilon > 0$ there is $T(\omega)$ with $1 - \pi_{t+1} < \varepsilon$ for $t \geq T$. Using $1 - \phi_t^{birth} \leq 1 - \phi_t$,

$$1 - \phi_{t+1} \leq \lambda(1 - \phi_t) + (1 - \lambda)\varepsilon, \quad t \geq T.$$

Iterating and taking limits gives $\limsup(1 - \phi_t) \leq \varepsilon$. Since ε is arbitrary, $\phi_t \rightarrow 1$ a.s.

Step 5. $\bar{n}_t \rightarrow n_R^*$ and $\sup_t \ln(N_t/\bar{N}) = +\infty$ a.s. $\square$

A.7 Proof of Proposition 3

Step 1: Learning bound. Linearizing (12) around π_t : $\pi_{t+1} - \pi_t \approx \pi_t(1 - \pi_t) \ln[\ell_R/\ell_M]$. Taking expectations under truth in the trap region ($|\ln(N_t/\bar{N})| < \delta$):

$$\mathbb{E}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] \leq \pi_t(1 - \pi_t)\mathcal{I}_t \leq \frac{1}{4}\bar{\mathcal{I}}(\delta) = C_1\bar{\mathcal{I}}(\delta).$$

Step 2: Selection bound.

$$\phi_t^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t} \leq \frac{(n_R - n_M)(1 - \phi_t)}{n_M},$$

giving $C_2 = 1/n_M$.

Step 3: Trap closure (induction on $\pi \leq \phi$). Suppose $(\phi_t, N_t, \pi_t) \in \mathcal{T}(\delta, \zeta)$, so $\pi_t \leq \phi_t$. Then $\pi_{t+1} - \phi_{t+1} = \pi_{t+1} - \lambda\phi_t^{birth} - (1 - \lambda)\pi_{t+1} = \lambda(\pi_{t+1} - \phi_t^{birth})$. Since $\phi_t^{birth} \geq \phi_t \geq \pi_t$ and (in the trap) $\pi_{t+1} - \pi_t$ is bounded above by $C_1\bar{\mathcal{I}}(\delta)$, which is small, we have $\pi_{t+1} \leq \pi_t + C_1\bar{\mathcal{I}}(\delta)$. For δ and ζ sufficiently small, this gives $\pi_{t+1} \leq \phi_{t+1}$, so trap condition (iii) propagates. The trap is closed under the joint dynamics.

Step 4: Drift bound and optional stopping.

$$\mathbb{E}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] \leq \lambda C_2(n_R - n_M)(1 - \phi_t) + (1 - \lambda)C_1\bar{\mathcal{I}}(\delta) \leq D,$$

where $D \equiv \lambda C_2(n_R - n_M) + (1 - \lambda)C_1\bar{\mathcal{I}}(\delta)$. Define $M_t = \phi_t - tD$; the bound shows $\mathbb{E}[M_{t+1} - M_t \mid \mathcal{F}_t] \leq 0$, so M_t is a supermartingale. By optional stopping,

$$\mathbb{E}[\phi_{T^{esc}}] - \phi_0 \leq D \cdot \mathbb{E}[T^{esc}],$$

giving $\mathbb{E}[T^{esc}] \geq (\bar{\phi} - \phi_0)/D$, which is the bound. $\square$

A.8 Trap Duration (Auxiliary Result)

Corollary 4. *Under Proposition 3, when learning is the binding escape channel, $\mathbb{E}[T^{esc}] = \Omega(1/\delta^2)$ as $\delta \rightarrow 0$.*

Proof. $\bar{\mathcal{I}}(\delta) = O(\delta^2)$. The lower bound $(\bar{\phi} - \phi_0)/(a + b\delta^2)$ with $a = \lambda C_2(n_R - n_M)$, $b = (1 - \lambda)C_1(\theta_R - \theta_M)^2/(2\sigma_\eta^2)$ gives $\Omega(1/\delta^2)$ when a is small relative to $b\delta^2$. $\square$

A.9 Trap Boundary (Auxiliary Result)

Proposition 8. *The critical population at which learning matches demographic selection is*

$$\frac{N^{crit}(\phi)}{\bar{N}} = \exp \left(\sqrt{\frac{2\sigma_\eta^2 \cdot \lambda C_2(n_R - n_M)(1 - \phi)}{(1 - \lambda)C_1(\theta_R - \theta_M)^2}} \right).$$

Proof. Equate the learning channel $(1 - \lambda)C_1(\theta_R - \theta_M)^2[\ln(N/\bar{N})]^2/(2\sigma_\eta^2)$ with the demographic selection channel $\lambda C_2(n_R - n_M)(1 - \phi)$ and solve. $\square$

A.10 Proof of Proposition 4

The proof is direct: a closed-form Bayes log-odds calculation for the learning channel, combined with a bounded comparison of the two drift terms.

Part (i): Demographic-selection drift. From $\bar{n}_t = \phi_t n_R + (1 - \phi_t)n_M$ and $\phi_t^{birth} = \phi_t n_R / \bar{n}_t$,

$$\phi_t^{birth} - \phi_t = \frac{\phi_t n_R - \phi_t \bar{n}_t}{\bar{n}_t} = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t},$$

which is strictly positive for any interior ϕ_t since $n_R > n_M$. The upper bound $(n_R - n_M)/(4n_M)$ follows from $\phi(1 - \phi) \leq 1/4$ and $\bar{n}_t \geq n_M$. This expression is independent of $\ln(N_t/\bar{N})$.

Part (ii): Bayesian learning drift. Define the log-odds $L_t = \ln[(1 - \pi_t)/\pi_t]$. Bayes' rule (12) gives

$$L_{t+1} - L_t = \ln \frac{\ell(s_t | \theta_M, N_t)}{\ell(s_t | \theta_R, N_t)}.$$

Substituting the Gaussian likelihoods (11) and simplifying:

$$L_{t+1} - L_t = \frac{(\theta_R - \theta_M) \ln(N_t/\bar{N})}{\sigma_\eta^2} \left[s_t - \bar{g} - \frac{\theta_R + \theta_M}{2} \ln(N_t/\bar{N}) \right].$$

Under truth $\theta^* = \theta_M$, $\mathbb{E}^{\theta_M}[s_t | \mathcal{F}_t] = \bar{g} + \theta_M \ln(N_t/\bar{N})$, so

$$\mathbb{E}^{\theta_M}[s_t - \bar{g} - \frac{\theta_R + \theta_M}{2} \ln(N_t/\bar{N}) | \mathcal{F}_t] = \frac{\theta_M - \theta_R}{2} \ln(N_t/\bar{N}).$$

Therefore

$$\mathbb{E}^{\theta_M}[L_{t+1} - L_t | \mathcal{F}_t] = \frac{(\theta_R - \theta_M) \ln(N_t/\bar{N})}{\sigma_\eta^2} \cdot \frac{\theta_M - \theta_R}{2} \ln(N_t/\bar{N}) = -\frac{(\theta_R - \theta_M)^2 [\ln(N_t/\bar{N})]^2}{2\sigma_\eta^2} = -\mathcal{I}_t,$$

which is the standard Kullback–Leibler identity for the Gaussian shift family. Since $\pi = 1/(1 + e^L)$ and $d\pi/dL = -\pi(1 - \pi)$, a first-order expansion gives

$$\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] = -\pi_t(1 - \pi_t)\mathcal{I}_t + o(\mathcal{I}_t),$$

which is strictly negative whenever $\pi_t \in (0, 1)$ and $\mathcal{I}_t > 0$, i.e., whenever $\ln(N_t/\bar{N}) \neq 0$.

Part (iii): Sign reversal of the joint drift. Combining (14) with parts (i) and (ii):

$$\mathbb{E}^{\theta_M}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] = \underbrace{\lambda \cdot \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t}}_{A(\phi_t) \geq 0, \text{ independent of } \ln(N_t/\bar{N})} - \underbrace{(1 - \lambda)\pi_t(1 - \pi_t)\mathcal{I}_t}_{B(\pi_t, N_t) \geq 0} + o(\mathcal{I}_t).$$

At $\ln(N_t/\bar{N}) = 0$, $\mathcal{I}_t = 0$ so $B = 0$ and the drift equals $A(\phi_t) > 0$ at any interior ϕ_t . As $|\ln(N_t/\bar{N})| \rightarrow \infty$, $\mathcal{I}_t \rightarrow \infty$ while $A(\phi_t) \leq \lambda(n_R - n_M)/(4n_M)$ remains bounded, so the drift becomes strictly negative for $|\ln(N_t/\bar{N})|$ sufficiently large. Equating $A = B$ at leading order, the threshold at which the drift changes sign solves

$$\bar{L}(\phi_t, \pi_t, \lambda)^2 = \frac{2\sigma_\eta^2 \cdot \lambda \phi_t(1 - \phi_t)(n_R - n_M)}{(1 - \lambda)\pi_t(1 - \pi_t)(\theta_R - \theta_M)^2 \bar{n}_t}.$$

Continuity of $\mathbb{E}^{\theta_M}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t]$ in $\ln(N_t/\bar{N})$ implies the existence of a state at which the drift vanishes and changes sign across this threshold; in particular, no monotone law in ϕ_t is consistent with the deterministic skeleton. $\square$

Remark on the interior steady state. Any candidate interior steady state with $\bar{n}^{ss} = 1$ requires the deterministic component of log-productivity growth to vanish: $\bar{g} + \theta_M \ln(N^{ss}/\bar{N}) = 0$, giving $\ln(N^{ss}/\bar{N}) = -\bar{g}/\theta_M > 0$ when $\theta_M < 0$. The interior steady state therefore lies at population strictly above the reference, and its location is uniquely determined up to the indeterminacy in ϕ^{ss} consistent with $\bar{n}^{ss} = 1$.

A.11 State-Dependent Value of Announcement (Auxiliary Results)

Proposition 9. *Suppose Assumption 4 holds and the economy is in $\mathcal{T}(\delta, \zeta)$ in a Romerian world. Under W^{tot} ,*

$$\mathcal{V}^{trap}(\tau_\nu^{trap}) = P^{esc}(\tau_\nu^{trap}) \cdot \Delta W^{esc} - P^{deepen}(\tau_\nu^{trap}) \cdot \Delta W^{deepen},$$

with a threshold $\underline{\tau}^{trap}$ above which $\mathcal{V}^{trap} > 0$. Because ΔW^{esc} is large, $\underline{\tau}^{trap}$ is moderate.

Proposition 10. *Suppose Assumption 4 holds. In a Malthusian world, where the dynamics are non-monotone by Proposition 4, under W^{tot} ,*

$$\mathcal{V}^{cycle}(\tau_\nu) = \Omega(\text{Var}(\theta_i) - \mathbb{E}[\text{Var}(\theta_i^{post})]) - \Lambda((\bar{\theta} - \theta^*)^2 - \mathbb{E}[(\bar{\theta}^{post} - \theta^*)^2]) - (\text{adjustment costs}),$$

where the second term captures the contribution of mean-bias correction to welfare and $\Lambda > 0$ is the appropriate curvature coefficient. For sufficiently precise signals, $\mathcal{V}^{cycle} > 0$, but the magnitude is smaller than $\mathcal{V}^{trap}$ when ΔW^{esc} is large.

Proof of Propositions 9 and 10. These follow from combining the welfare decomposition (26) with the state-dependent value of coordination. In the trap, the gain from escape ΔW^{esc} is large, so even a moderately noisy announcement is valuable if it has positive probability of triggering escape. In the non-monotone regime of Proposition 4, the gain comes from reducing dispersion (the Δ_3 channel) and mean bias (the Δ_2 channel), while adjustment costs from disrupting the existing dynamics partially offset benefits. The threshold $\underline{\tau}^{trap}$ is determined by equating expected gain and expected loss. $\square$

B Welfare Analysis Proofs and Auxiliary Results

B.1 Proof of Proposition 5

Step 1: Planner's Bellman. The planner's problem (24) has Bellman

$$V^P(A_t, N_t) = \max_{n_t} \{N_t[\ln A_t + \ln(1 - \psi(n_t))] + \beta \mathbb{E}^* V^P(A_{t+1}, n_t N_t)\},$$

with $\ln A_{t+1} = \ln A_t + \bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}$. Conjecture (in the local log-linear sense) $V^P(A, N) = N \cdot w^P(A, N)$ with $w^P \approx v^P + \xi_A^P \ln A + \xi_N^P \ln(N/\bar{N})$. Substituting and dividing by N_t :

$$w^P(A_t, N_t) = \ln A_t + \ln(1 - \psi(n_t^*)) + \beta n_t^* [v^P + \xi_A^P (\ln A_t + \bar{g} + \theta^* \ln(N_t/\bar{N})) + \xi_N^P (\ln(N_t/\bar{N}) + \ln n_t^*)] + \text{const.}$$

Matching coefficients at the reference: $\xi_A^P = 1/(1 - \beta n_t^*)$ and $\xi_N^P = \beta n_t^* \theta^*/(1 - \beta n_t^*)^2$.

Step 2: Planner FOC. The marginal value of population is $V_N^P(A', N') = w^P(A', N') + \xi_N^P$. The FOC gives (25).

Step 3: Comparison with the household. The household's common-correct-belief FOC has the same form but *without* ξ_N^P . The wedge is $\beta \xi_N^P$, with sign θ^* . With strictly increasing LHS:

- $\theta^* > 0$: $\xi_N^P > 0$, so $n^{FB} > n^{CB}$.
- $\theta^* < 0$: $\xi_N^P < 0$, so $n^{FB} < n^{CB}$.

Step 4: Belief distortion. $n_M^* < n^{CB}$ by Lemma 1, so in a Romerian world, $n^{FB} > n^{CB} > n_M^*$. $\square$

B.2 First-Best under Average Utilitarianism (Auxiliary Result)

Proposition 11. *Let $n^{FB,tot}$ and $n^{FB,avg}$ denote first-best fertility under total and average utilitarianism. In a Romerian world, $n^{CB} < n^{FB,avg} < n^{FB,tot}$. The average-utilitarian planner internalizes the per-capita scale-effect channel through $\ln A_t$, giving wedge $\beta \xi_{N,avg}^P > 0$, but lacks the population-multiplier term N_t that gives the total utilitarian planner the additional wedge $\beta(\xi_N^P - \xi_{N,avg}^P) > 0$.*

Proof sketch. The average-utilitarian Bellman is $V^{P,avg}(A, N) = \max_n \{[\ln A + \ln(1 - \psi(n))] + \beta \mathbb{E}^* V^{P,avg}(A', N')\}$, with no population multiplier. Conjecturing $V^{P,avg}(A, N) \approx v^{P,avg} + \xi_A^{P,avg} \ln A + \xi_N^{P,avg} \ln(N/\bar{N})$ and matching coefficients at the reference yields $\xi_A^{P,avg} = 1/(1 - \beta)$ (now without an n^* feedback in the denominator) and $\xi_N^{P,avg}$ carrying the same θ^* sign as ξ_N^P but strictly smaller in magnitude because the population-multiplier term N_t in the total utilitarian objective amplifies the marginal value of population. The FOC ordering follows. $\square$

B.3 Proof of Proposition 6

Step 1: Define benchmarks. $W^{FB}(\theta^*)$, $W^{CB}(\theta)$ (common-belief decentralized welfare when all households believe θ), W^{HB} .

Step 2: Decomposition.

$$W^{FB}(\theta^*) - W^{HB} = [W^{FB}(\theta^*) - W^{CB}(\theta^*)] + [W^{CB}(\theta^*) - W^{CB}(\bar{\theta})] + [W^{CB}(\bar{\theta}) - W^{HB}] = \Delta_1 + \Delta_2 + \Delta_3.$$

Step 3: Sign of Δ_1 . $\Delta_1 \geq 0$ since the planner can always replicate the common-belief decentralized fertility, with equality iff $\xi_N^P = 0$, i.e., $\theta^* = 0$. Δ_1 is increasing in $|\theta^*|$ by direct calculation.

Step 4: Sign of Δ_2 . $W^{CB}(\theta)$ is maximized at $\theta = \theta^*$ under truth-evaluation. Second-order Taylor expansion gives $\Delta_2 \approx \frac{1}{2} W^{CB, \prime\prime}(\theta^*)(\bar{\theta} - \theta^*)^2$; the leading coefficient is positive.

Step 5: Δ_3 . Take Taylor expansion around the social-optimum fertility n^{SP} where the externality is internalized:

$$W^{CB}(\bar{\theta}) - W^{HB} \approx \frac{\omega \kappa^2}{2} \text{Var}(\theta_i),$$

with $\omega = -V''(n^{CB}) > 0$ and κ the slope of the fertility policy in beliefs. The first-order externality term has been absorbed into Δ_1 by construction. $\square$

B.4 Lemma: Welfare Loss from Belief Dispersion

Lemma. *Aggregate (true-evaluation) welfare under heterogeneous beliefs satisfies*

$$W^{CB}(\theta^*) - W^{HB} \approx \frac{\omega \kappa^2}{2} [\text{Var}(\theta_i) + (\bar{\theta} - \theta^*)^2].$$

Proof. $W^{HB} = \phi V_R + (1 - \phi) V_M$. Quadratic expansion: $V^* - V_j \approx \frac{\omega}{2} (n_j^* - n^{CB})^2$ with $\omega = -V''(n^{CB}) > 0$. Linearization: $n_j^* - n^{CB} \approx \kappa(\theta_j - \theta^*)$. Combining and using $\mathbb{E}_i[(\theta_i - \theta^*)^2] = \text{Var}(\theta_i) + (\bar{\theta} - \theta^*)^2$ gives the result. $\square$

B.5 Proof of Theorem 2

Step 1. $W^{dyn} = \phi V_R + (1 - \phi) V_M$ under subjective beliefs. The announcement μ shifts subjective posteriors via (28).

Step 2: Blackwell sufficiency. Under any prior, an information signal cannot reduce expected utility (Blackwell). The signal structure $z = \mu + \tilde{\nu}$ with $\tilde{\nu} \sim \mathcal{N}(0, \tau_A^{-1})$ delivers a posterior on $\{\theta_R, \theta_M\}$ whose informativeness about θ^* (as measured by mutual information) is maximized when $\mu = \mathbb{E}[\theta^* \mid \text{planner's information}] = \hat{\theta}$. Any deterministic deviation $\mu \neq \hat{\theta}$ introduces systematic bias relative to the planner's posterior and strictly reduces the Blackwell informativeness of the announcement.

Step 3: No incentive to distort. Distortion strictly reduces dynastic welfare under subjective preferences. Hence $\mu^{dyn} = \hat{\theta}$, with maximal precision $\sigma_A^{dyn} = 0$ (point mass) preferred over noisier signal structures. $\square$

B.6 Proof of Theorem 3

Step 1. $W^{tot}(\mu)$ under truth, with μ shifting subjective posteriors and hence equilibrium fertility.

Step 2: First-order condition.

$$\frac{dW^{tot}}{d\mu} = \underbrace{\sum_j \omega_j \frac{\partial V_j^{tot}}{\partial \theta_j^{post}} \cdot \frac{\partial \theta_j^{post}}{\partial \mu}}_{\text{(belief correction)}} + \underbrace{\frac{\partial W^{tot}}{\partial \bar{n}} \bigg|_{\bar{n}^*} \cdot \frac{d\bar{n}^*}{d\mu}}_{\text{(externality)}}.$$

The first term vanishes at $\mu = \hat{\theta}$ (the dynastic-truthful benchmark). The second is positive in a Romerian world because the planner internalizes the wedge $\beta \xi_N^P > 0$.

Step 3: Optimum. Setting the FOC to zero gives $\mu^{tot} = \hat{\theta} + \chi^{tot}$. To leading order, χ^{tot} solves $\chi^{tot} \cdot \omega \kappa^2 \tau_A \approx \beta \xi_N^P \cdot \kappa \tau_A$, so

$$\chi^{tot} \approx \frac{\beta \xi_N^P}{\omega \kappa} > 0 \quad \text{if } \theta^* > 0.$$

Step 4: Comparative statics. (i) $\partial \chi^{tot} / \partial |\theta^*| > 0$: ξ_N^P is monotone in θ^* . (ii) $\partial \chi^{tot} / \partial |\ln(N_0/\bar{N})| > 0$: when population is far from $\bar{N}$, the externality wedge enters welfare scaled by N_0 and is larger in absolute terms. (iii) $\partial \chi^{tot} / \partial \text{Var}(\theta_{i,0}) < 0$: high-variance priors are themselves moved more by the announcement, so the marginal benefit of additional shading falls. (iv) $\partial \chi^{tot} / \partial \tau_A > 0$: more credible announcements pass through more strongly to fertility, raising the marginal benefit of shading.

Step 5: Sign reversal. When $\theta^* < 0$, $\xi_N^P < 0$ and $\chi^{tot} < 0$: the planner shades toward pessimism. $\square$

B.7 Proof of Corollary 1

The argument follows the structure of Theorem 3 with W^{avg} replacing W^{tot} . The belief-correction term again vanishes at $\mu = \hat{\theta}$. The externality term retains the same sign as in the total utilitarian case but uses the smaller wedge $\beta \xi_{N,avg}^P$ from Proposition 11 in place of $\beta \xi_N^P$. To leading order,

$$\chi^{avg} \approx \frac{\beta \xi_{N,avg}^P}{\omega \kappa},$$

which has the same sign as χ^{tot} (matching that of θ^*) but strictly smaller magnitude. Hence $0 < \chi^{avg} < \chi^{tot}$ in a Romerian world, with the inequalities reversing in a Malthusian world. $\square$

B.8 Proof of Corollary 2

The first equality is Theorem 2. The middle inequality is Corollary 1 together with $\chi^{avg} > 0$ in a Romerian world. The right inequality is $\chi^{avg} < \chi^{tot}$, again from Corollary 1. The Malthusian-world reversal follows from the corresponding sign reversals of the wedges. $\square$